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Asymmetric synthesis

Connections

➡ Building on

- Carbonyl group reactions **ch6, ch9–ch11**
- Stereochemistry and conformation **ch14, ch16, & ch31**
- Electrophilic addition to enolates and alkenes **ch19 & ch20**
- Aldol reactions **ch26**
- Diastereoselectivity **ch32 & ch33**
- Cycloadditions **ch34**

Arriving at

- Why making pure enantiomers matters
- Chirality derives from nature
- The chiral pool provides starting materials, auxiliaries, and catalysts
- Chiral auxiliaries work well in asymmetric alkylation and aldol reactions
- Chiral catalysts for oxidation and reduction reactions
- Ligand-accelerated catalysis
- Catalysis with and without metals

➡ Looking forward to

- Chemistry of life **ch42**
- Chemistry and the future **ch43**



'L'univers est dissymétrique'.
Louis Pasteur, *Comptes Rendus Acad. Sci.*, Paris June 1, 1874.

■ This chapter builds on the concepts introduced in Chapter 14: make sure you understand all the terms used to describe stereochemistry that are defined there. In particular make sure you are absolutely clear on the meanings of *chiral*, *achiral*, *enantiomer*, and *diastereoisomer*, along with what the designators *R*, *S*, *+*, *–*, *L*, and *D* refer to.

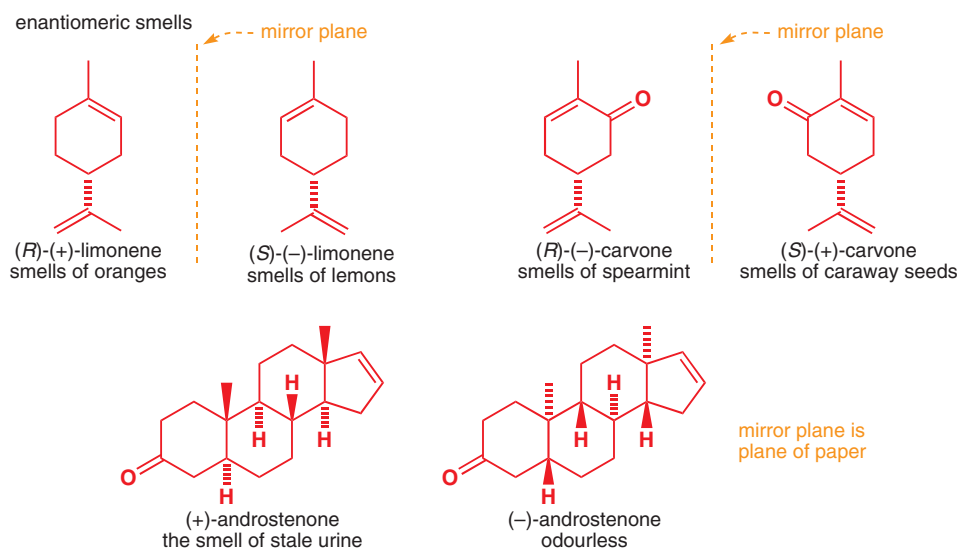
Nature is asymmetric

'How would you like to live in Looking-glass House, Kitty? I wonder if they'd give you milk in there? Perhaps looking-glass milk isn't good to drink. . .' Lewis Carroll, *Through the looking-glass and what Alice found there*, Macmillan, 1872.

You are chiral, and so are Alice, Kitty, and all living organisms. You may think you look fairly symmetrical in a looking-glass, but as you read this book you are probably turning the pages with your right hand and processing the information with the left side of your brain. Some organisms are rather more obviously chiral: snails, for example, carry shells that could spiral to the left or to the right. Not only is nature chiral, but by and large it exists as just one enantiomer—although some snail shells spiral to the left, the vast majority of marine snail shells spiral to the right; humans have their stomach on their left and their liver on their right; honeysuckle (*Lonicera*) climbs by spiralling to the left and all bindweed (*Convolvulus*) spirals to the right.

Nature has a left and a right, and it can tell the difference between them. You may think that human beings are sadly lacking in this respect, since as children we all had to learn, rather laboriously, which is which. Yet at an even earlier age, you could no doubt distinguish the smell of oranges from the smell of lemons, even though this is an achievement at least as remarkable as getting the right shoe on the right foot. The smells of orange and lemon differ in being the left- and right-handed versions of the same molecule, limonene. (*R*)-(+)-Limonene smells rounded and orangey; (*S*)-(–)-limonene is sharp and lemony. Similarly, spearmint and caraway seeds smell quite different, although again this pair of aromas differs only in being the enantiomeric forms of the ketone carvone. Evolution has left many of us regrettably sensitive to (+)-androstene, the smell of stale human urine. (–)-Androstene is essentially odourless.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1, 123-2** etc. (replacing **123** with the page number) for access to successive links.

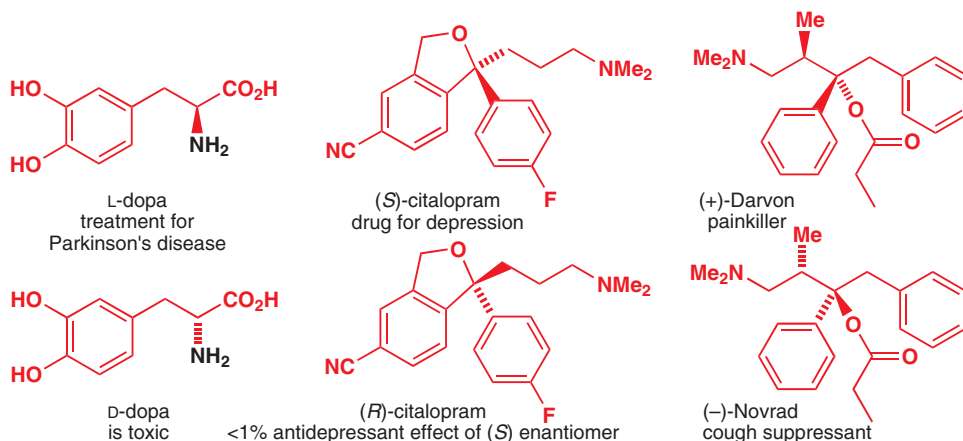


Even bacteria know their right from their left: *Pseudomonas putida* can use aromatic hydrocarbons as a foodstuff, degrading them to diols. The diol produced from bromobenzene is formed as one enantiomer only.

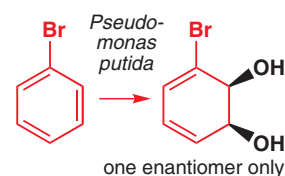
How can this be? We said in Chapter 14 that enantiomers are chemically identical, so how is it that we can distinguish them with our noses and bacteria can produce them selectively? Well, the answer lies in a proviso to our assumption about the identity of enantiomers: they are identical *until they are placed in a chiral environment*. This concept will underlie all we say in this chapter about how to make single enantiomers in the laboratory. We take our lead from nature: all life is chiral, so all living systems are chiral environments.

The sheer complexity of life means that nature has to build its living structures from molecules that are chiral, principally amino acids and sugars. For all of those chiral molecules, evolution has forced the use of a single enantiomeric form, for example every amino acid in your body has the same configuration (usually labelled *S*). From this fact derives the larger-scale chirality of all living structures, from the right-handed double helix of DNA to the location of a blue whale's internal organs. The answer to the question posed by Alice at the start of the chapter is most certainly *no*—her kitten's digestive system will be able to hydrolyse the achiral fats in the looking-glass milk quite easily (achiral compounds are superimposable on their mirror image), but looking-glass proteins (which will be made of *D*-amino acids) and *L*-lactose will be quite indigestible.

For a perfumer or flavour and fragrance manufacturer, the distinction between the differently scented enantiomers of the same molecule is clearly of great importance. Nonetheless, we could all get by with caraway-flavoured toothpaste. Yet when it comes to drug molecules, making the right enantiomer can be a matter of life and death. Parkinson's disease sufferers are treated with the non-proteinogenic amino acid dopa (3-(3,4-dihydroxyphenyl)alanine). Dopa is chiral, and only (*S*)-dopa (known as *L*-dopa) is effective in restoring nerve function. (*R*)-Dopa is not only ineffective, it is quite toxic, so the drug must be marketed as a single enantiomer.

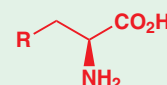


(+)-Androstrenone is also a pig pheromone. You may not wish to know that it is the active component of DuPont's Boarmate, used by pig farmers to prepare sows for artificial insemination.

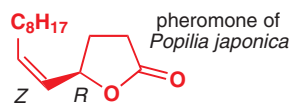


➡ The molecules of life are examined in detail in the next chapter.

■ Natural *L*-amino acids all look like this:



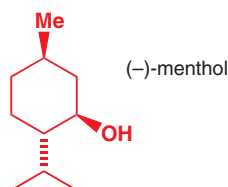
All have *S* stereochemistry except cysteine ($R=SH$), where the priority rules mean the chiral centre is *R*. Some bacteria make their cell walls from 'unnatural' *R*-amino acids to make them unassailable by the (*S*-amino acid-derived) enzymes used by higher organisms to hydrolyse peptides (see p. 1141).



In other cases, only one of the two enantiomers of a drug molecule possesses activity: the antidepressant citalopram and the painkiller naproxen are both marketed only as their *S* enantiomer because the *R* enantiomers are essentially inactive. In a few cases, the enantiomers both have activity, but in different ways: (+)-Darvon and (–)-Novrad are a painkiller and a cough suppressant, respectively.

It is not only drugs that have to be manufactured enantiomerically pure. This simple lactone is the pheromone released by the Japanese beetle *Popilia japonica* as a means of communication. The beetles, whose larvae are serious crop pests, are attracted by the pheromone, and synthetic pheromone is marketed as 'Japonilure' to bait beetle traps. Provided the synthetic pheromone is the stereoisomer shown, with the *Z* double bond and the *R* configuration at the stereogenic centre, only 25 µg per trap catches thousands of beetles. You met this compound in Chapter 27, where we pointed out that double bond stereocontrol is important since the *E* isomer of the pheromone is virtually useless as a bait (it retains only about 10% of the activity). Even more important is control over the configuration at the chiral centre because the *S* enantiomer of the pheromone is not only inactive in attracting the beetles, but acts as a powerful inhibitor of the *R* enantiomer—even 1% of *S* enantiomer in a sample of pheromone destroys the activity.

So you see why chemists need to be able to make compounds as single enantiomers. In Chapters 32 and 33 we looked at *relative stereochemistry* and how to control it; this chapter is about how to control *absolute stereochemistry*. We call this asymmetric synthesis.

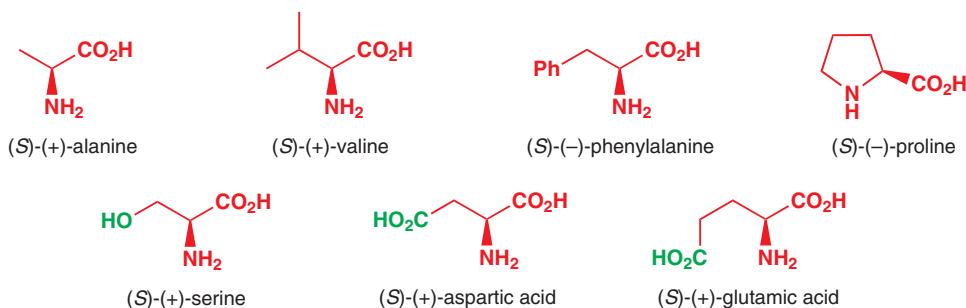


In the last 25 years or so, this subject has occupied more organic chemists than possibly any other, and we are now at a point where it is not only possible (and in fact essential because of strict regulatory rules) to make many drug molecules as single enantiomers, but it is also even possible to make many chiral molecules that are indigenous to nature more cheaply in the laboratory. By 2007, for example, at least 30% of the world's supply of menthol was not extracted from plants but made synthetically. A thousand tonnes of (–)-menthol a year is made by the company Takasago in Japan using the techniques of asymmetric synthesis that you will meet later in this chapter.

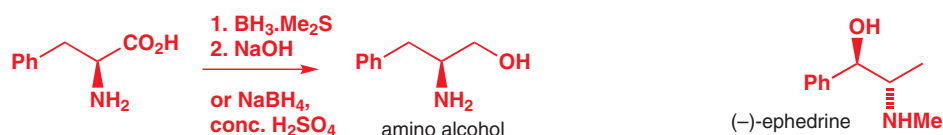
The chiral pool: Nature's chiral centres 'off the shelf'

When we first introduced you to enantiomers and chirality in Chapter 14, we stressed that any imbalance in enantiomers always derives ultimately from nature. A laboratory synthesis of a chiral compound from achiral or racemic starting materials alone always gives a racemic mixture of enantiomers. If you want to make just one enantiomer, you have to use a starting material or reagent which is also just one enantiomer. This seems like a chicken-and-egg situation, until you realize that nature provides a collection of 'off the shelf' enantiomerically pure compounds that we can exploit in various ways. This collection of natural, enantiomerically pure compounds is called the **chiral pool**. The principal groups of compounds in the chiral pool are:

1. The amino acids. There is a full list of the natural amino acids found in proteins on p. 554, but for the purposes of this chapter you should make sure you are familiar with the structures below. They all have simple side chains that are simple alkyl groups or functionalized chains with plenty of versatile chemistry, and can be obtained by hydrolysis of protein.



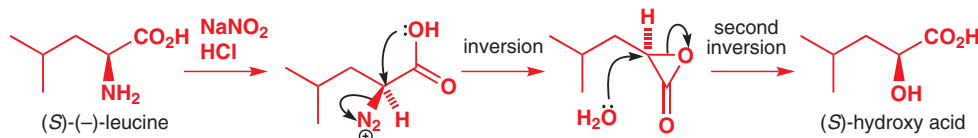
2. Simple derivatives of the amino acids: amino alcohols and hydroxy acids. It's easy to reduce amino acids to amino alcohols with borane (BH_3), usually generated in the reaction mixture by treating sodium borohydride with concentrated sulfuric acid. We will use a number of naturally derived amino alcohols as starting materials in this chapter.



Ephedrine is an amino alcohol which is itself a useful member of the chiral pool—it's a plant extract readily available as either diastereoisomer (see p. 314), each, unusually, also available as either enantiomer.

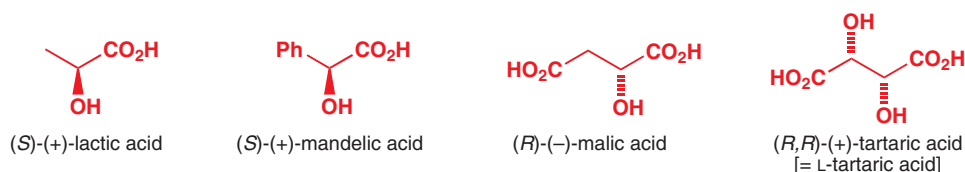
It's also easy to make hydroxy acids from amino acids by diazotization. You saw this being done in Chapter 33, but as a reminder nitrous acid generates a diazonium salt, which undergoes substitution by water via an intermediate α -lactone. Two configurational inversions are involved, so the product alcohol retains *S* stereochemistry.

diazotization–hydrolysis of amino acids to give hydroxy acids proceeds with overall retention



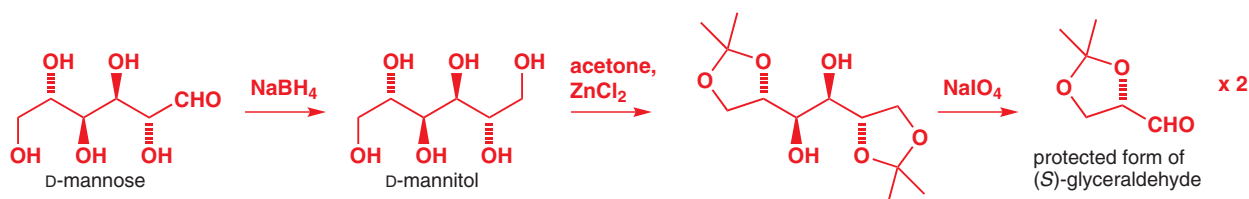
➡ Look back at p. 875 for more on the mechanism of this transformation—it's very important that the reaction goes with overall *retention* of stereochemistry.

Some hydroxy acids are themselves available from nature, and are therefore also members of the chiral pool: both (*R*)- and (*S*)-lactic acid, for example, can be made by bacterial fermentation; mandelic, malic, and tartaric acids are extracted from almonds, apples, and grapes, respectively.



3. Carbohydrates and their derivatives. There are a great many simple carbohydrates available, but one of the most useful is mannose. Reduction to the alcohol gives the C_2 -symmetric compound mannitol, which can be converted to a useful aldehyde by selective protection as a bis-acetal with acetone and a Lewis acid. Cleavage of the remaining diol with sodium periodate gives two equivalents of a useful protected form of glyceraldehyde.

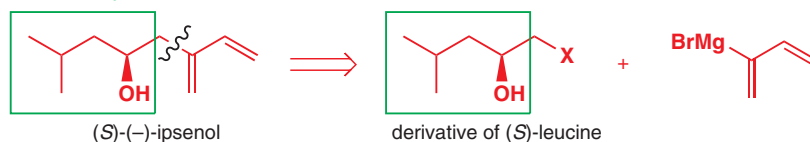
➡ The concept of C_2 rotational symmetry is discussed on p. 320. C_2 symmetry is compatible with chirality. For the oxidative cleavage of diols to aldehydes, see p. 443.



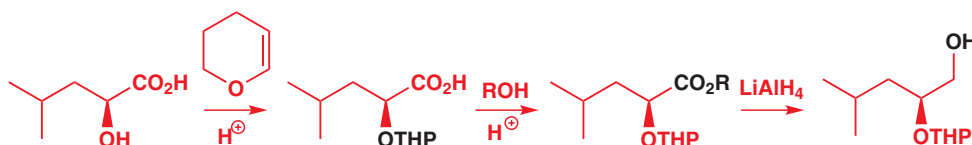
In this chapter we will show you the many and varied ways in which members of the chiral pool can be put to work in asymmetric synthesis, but the most straightforward application is simply to spot that a target molecule has a close structural similarity with, say, an amino

acid. This is what Mori did when he made another important insect pheromone, ipsenol. The left-hand half of the molecule has the same structure as the side chain of leucine, and the *S* chiral centre can also come from (*S*)-leucine.

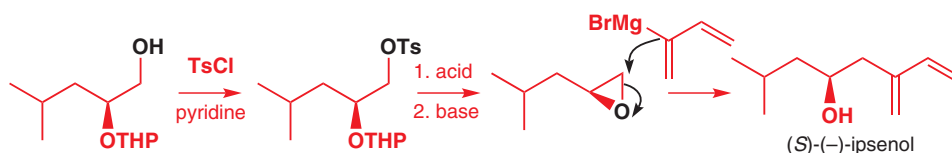
structural similarity with leucine



Mori used (*S*)-leucine as the starting material and converted it to the (*S*)-hydroxy acid by the method on p. 875. The hydroxyl group was protected as the THP derivative (Chapter 23).



Reduction of the acid, via the ester, then allowed introduction of the tosylate leaving group, which was displaced to make an epoxide. The epoxide was opened by a Grignard reagent to introduce the diene portion and give the target molecule.

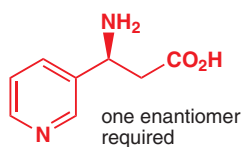


➡ This drawback is highlighted in the synthesis of oseltamivir in Chapter 43 (p. 1174).

This might seem rather long-winded, and long-windedness can be a drawback of syntheses starting from the chiral pool: you have to shoe-horn your synthetic route into the available starting materials. Another drawback of syntheses starting from the chiral pool is the fact that many natural compounds are only available as one enantiomer or, if both enantiomers are available, one is much more expensive than the other. You will see some ingenious ways of circumventing this problem later in the chapter, but we deal with a very simple one in the next section.

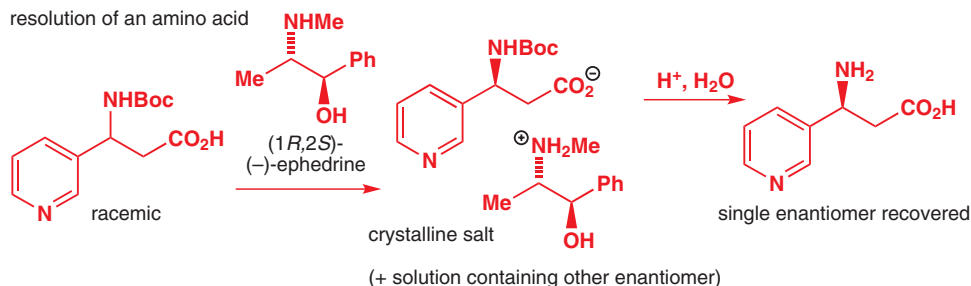
Resolution can be used to separate enantiomers

➡ We will not explain resolution again here: turn to p. 322 for the details.



In Chapter 14 we introduced you to resolution as a means of separating enantiomers. Resolution requires an enantiomerically pure resolving agent, which must be a compound from the chiral pool or a simple derivative of that compound. When the Swiss company Cilag wanted one enantiomer of the unusual chiral amino acid in the margin in order to make some potential drug candidates, the chemists there decided the easiest way to get hold of it quickly and in large quantities was to make it in racemic form and then resolve it. It turned out that one of the two enantiomers of the protected derivative below forms a crystalline salt with cheap, readily available (–)-ephedrine, while the other remains in solution. Filtration and treatment with acid to remove the protecting group and protonate the acid gave them a single enantiomer of their target amino acid.

resolution of an amino acid



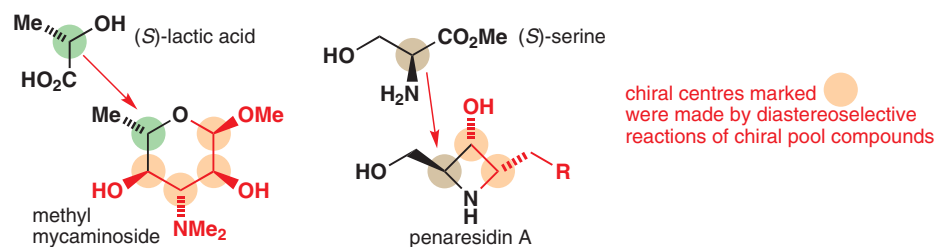
Of course, with resolution, there is a maximum yield of 50% because if you only want one enantiomer, the other is wasted. But there are many cases where you might want *both* enantiomers. You may need to test them both for biological activity, for example. In that case, resolution is ideal—in the example above the chemists at Cilag could get hold of the other enantiomer of the amino acid just by evaporating the mother liquors from the recrystallization. This is a big advantage of resolution: it lets you get both enantiomers using just one compound from the chiral pool.

Chiral auxiliaries

In Chapter 33 we showed you methods for making single diastereoisomers using diastereoselective reactions. Diastereoselective reactions work just as well whether the starting material is racemic or enantiomerically pure—you get the same diastereoisomeric outcome in each case, but if you start with racemic material you get racemic product and if you start with enantiomerically pure material you get enantiomerically pure product. Here's an example, from p. 867:



So if you use a starting material from the chiral pool, you can build new chiral centres in enantiomerically pure form just by using diastereoselective reactions. We showed you two syntheses at the end of Chapter 33 using this idea: the chiral pool starting materials (*S*)-lactic acid and (*S*)-serine were converted to two natural products using a series of diastereoselective reactions to introduce further chiral centres into the molecules.



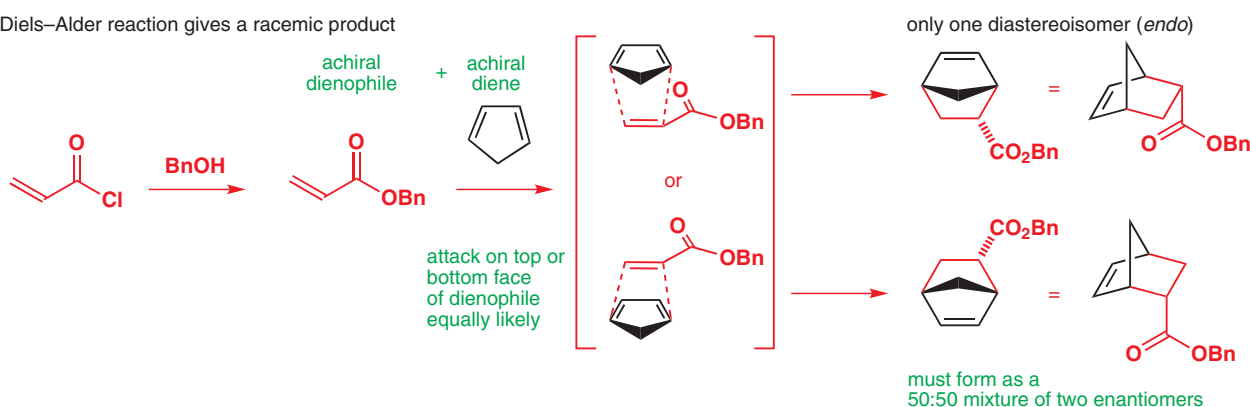
➡ These syntheses are on pp. 872–875.

The syntheses rely on the fact that the structure of the chiral pool starting material is still there in the product. But the same idea can work even if the starting chiral compound is no longer part of the target you are making. In this case the chiral starting material is called a *chiral auxiliary*. Chiral auxiliaries are extremely versatile because they can be used to make a whole variety of target molecules in enantiomerically pure form. We will explain how they work with two examples.

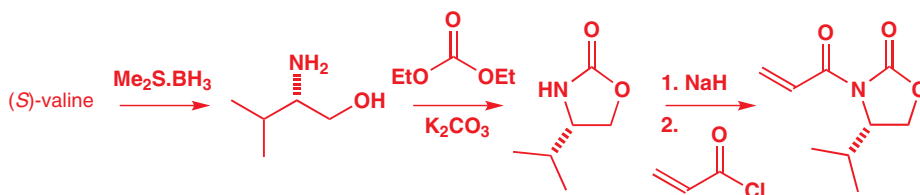
➡ Diels–Alder reactions of cyclopentadiene appear on p. 880.

The product of a Diels–Alder reaction between cyclopentadiene and benzyl acrylate must necessarily be racemic as both reagents are achiral. Although only one *diastereoisomer*—the *endo* product—is formed, it must be formed as an exactly 50:50 mixture of *enantiomers*. There is nothing to tell the diene whether to attack the top or the bottom face of the dienophile so it does both, each 50% of the time.

Diels–Alder reaction gives a racemic product

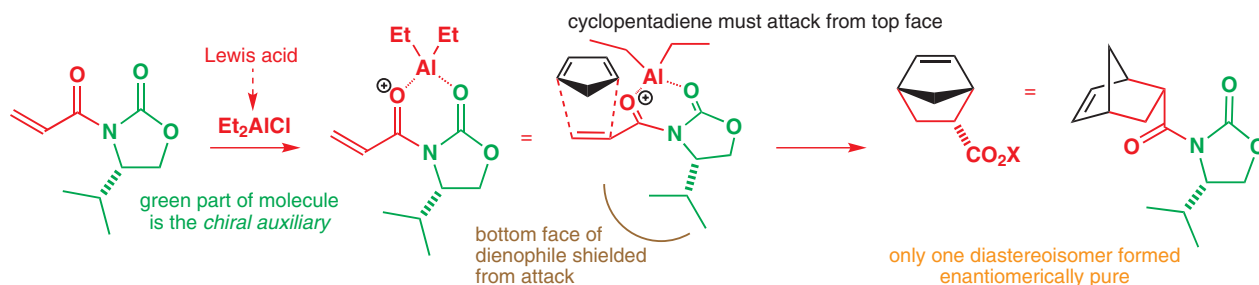


Now see what happens if we replace the achiral benzyl ester group of the dienophile with an amide derived from the amino acid valine. Here's the synthesis of such a dienophile using the amino acid reduction you saw on p. 1105.



■ As we discussed in Chapter 34, the presence of a Lewis acid increases the rate of Diels–Alder reactions, and in this case is also vital for high stereoselectivity.

The two faces of the double bond of the dienophile are now different because of the chiral centre: they are diastereotopic, and the diene can distinguish between them. If we now do the Diels–Alder reaction in the presence of a Lewis acid, Et_2AlCl , the aluminium chelates the oxygen atoms of the dienophile to form the rigid and reactive structure shown below. The isopropyl group is held in such a way that its steric bulk prevents the diene attacking that face of the prochiral alkene. The diene has no choice but to attack from above, and only one of the possible diastereoisomeric products is formed.

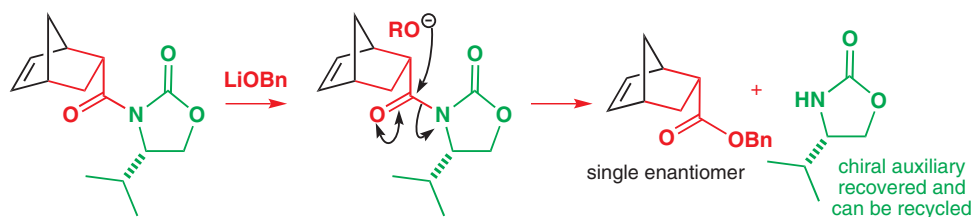


Interactive chiral auxiliary-controlled Diels–Alder reaction

We call the green valine-derived part of this molecule the *chiral auxiliary*—it assists the substrate to react in a diastereoselective way such that only one of the two possible products is allowed to form. The chiral auxiliary was enantiomerically pure to start with, so the product must be diastereoisomerically *and* enantiomerically pure.

Finally comes the step which shows the power of chiral auxiliary strategy: we just remove the chiral auxiliary from the product by treating with a nucleophile. The auxiliary can in principle be used again, but most importantly of all, the product obtained is just one of the two enantiomers we made in the racemic version of this reaction. This isn't a resolution—all

of these steps go in high yield—it is truly an enantioselective synthesis of the Diels–Alder product, using a chiral auxiliary to help us.



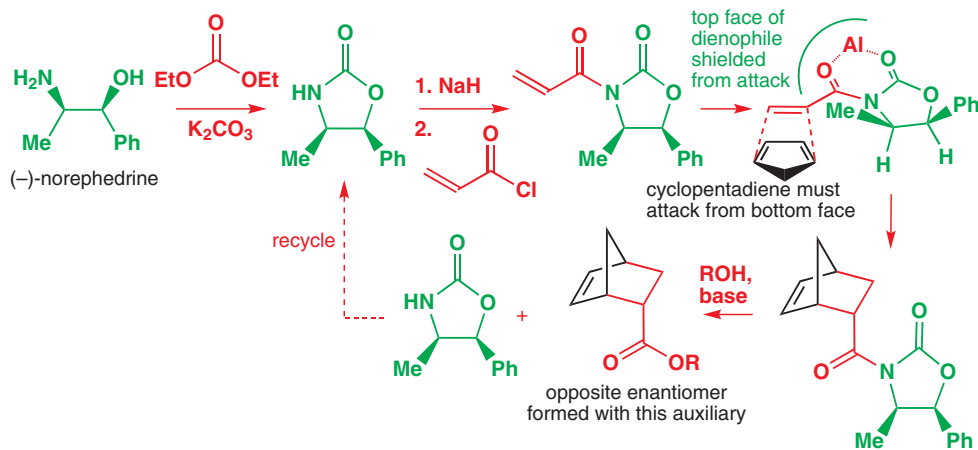
Overall, by sequential attachment of the auxiliary, diastereoselective reaction, and removal of the auxiliary we have made the same product but as a single enantiomer.

● **This is what we mean by a chiral auxiliary strategy**

- 1 An enantiomerically pure compound (usually derived from a simple natural product like an amino acid), called a chiral auxiliary, is attached to the starting material.
- 2 A diastereoselective reaction is carried out, which, because of the enantiomeric purity of the chiral auxiliary, gives only one enantiomer of the product.
- 3 The chiral auxiliary is removed by, for example, hydrolysis, leaving the product of the reaction as a single enantiomer. The best chiral auxiliaries (of which the example above is one) can be recycled, so although stoichiometric quantities are needed, there is no waste.

We have introduced you to this chiral auxiliary before any other because it is more commonly used than any other. It is a member of the oxazolidinone (the name of the heterocyclic ring) family of auxiliaries developed by David Evans at Harvard University, and is easily and cheaply made from the amino acid (*S*)-valine. Even though it is cheap, it can be recycled. The last step of the route above regenerates the auxiliary ready for re-use.

The most versatile chiral auxiliaries should also be available as both enantiomers. For the valine-derived one here, this is not the case—(*R*)-valine is quite expensive since it is not found in nature. However, by starting with the naturally occurring (and cheap) compound nor-ephedrine, we can make an auxiliary that, although not enantiomeric with the one derived from (*S*)-valine, acts as though it were.



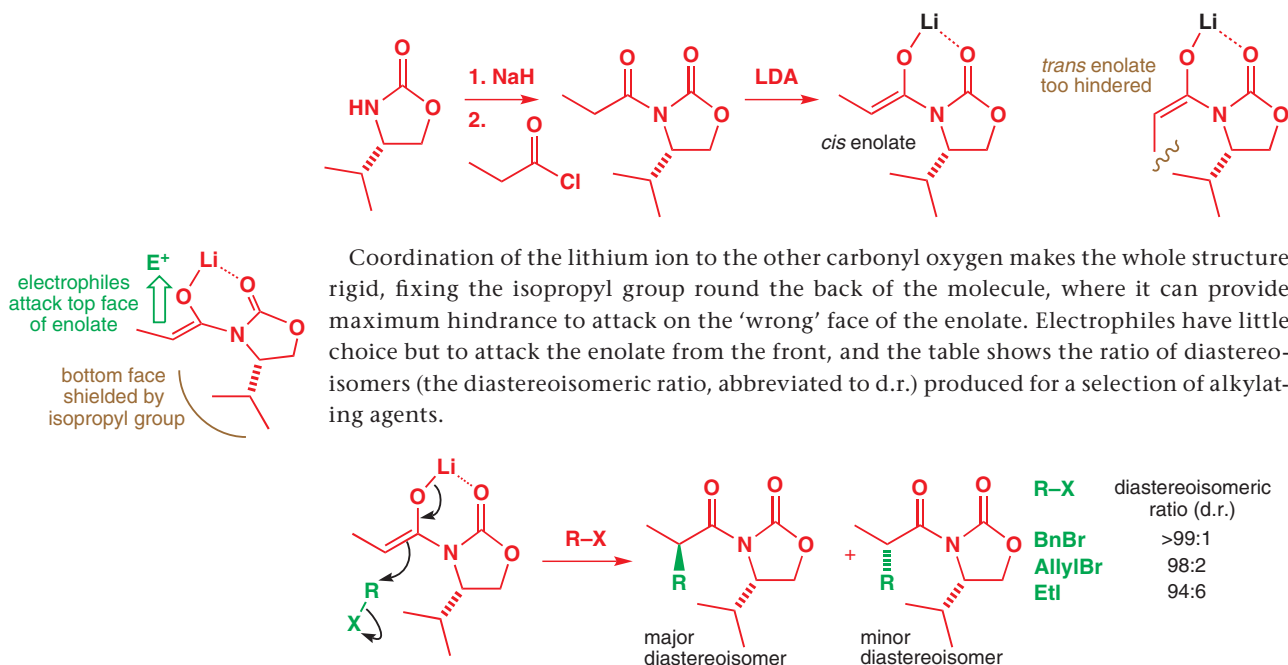
As the diagram shows, the two substituents of the auxiliary lie on the top face of the dienophile and force the cyclopentadiene this time to attack the bottom face. Now when the auxiliary is cleaved from the product the opposite enantiomer is formed. We can choose which enantiomer we want simply by choosing the right auxiliary for the job.

Alkylation of enolates

Chiral auxiliaries can be used in plenty of other reactions, and some of the most common are reactions of enolates. Evans's oxazolidinone auxiliaries are particularly appropriate here

➡ Enolates are a sort of alkene and can form as *cis* or *trans* geometrical isomers. One of the consequences of this is discussed in Chapter 33.

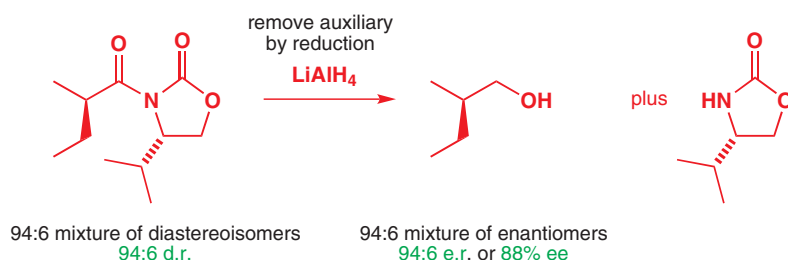
because they are readily turned into enolizable carboxylic acid derivatives. Treatment with base (usually LDA) at low temperature produces an enolate, the bulky auxiliary means that only the *cis* enolate forms: the *trans* enolate is too hindered.



As you can see, none of these reactions is truly 100% diastereoselective and, indeed, only the best chiral auxiliaries (of which this is certainly one) give >98% of a single diastereoisomer. The problem with less than perfect diastereoselectivity is that, when the chiral auxiliary is removed, the final product is contaminated with some of the other enantiomer. A 94:6 ratio of diastereoisomers will result in a 94:6 ratio of enantiomers, or a sample of 94:6 e.r. (e.r. for enantiomeric ratio).

Enantiomeric excess

Compounds that are neither racemic nor enantiomerically pure are usually called *enantiomerically enriched*. Chemists have two ways of referring to the ratio of enantiomers in an enantiomerically enriched sample. The first is the simple one we have just used: e.r. or enantiomeric ratio, expressed as two numbers adding to 100. More common, however, is to express this ratio as an *enantiomeric excess*. Enantiomeric excess (or ee) is defined as the excess of one enantiomer over the other, expressed as a percentage of the whole. So a 94:6 mixture of enantiomers consists of one enantiomer in 88% excess over the other, and we call it an enantiomerically enriched mixture with 88% ee. Why not just say that we have 94% of one enantiomer? Enantiomers are not like other isomers because they are simply mirror images. The 6% of the minor enantiomer can be paired with 6% of the major isomer to form a racemic mixture amounting to 12% of the total. The mixture contains 12% racemate and 88% of one enantiomer, hence 88% ee.



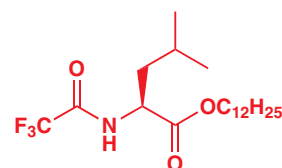
We will see shortly how we can make further use of the chiral auxiliary to increase the ee of the reaction products. But first, we should consider how to measure ee. One way is simply to measure the angle through which the sample rotates plane-polarized light. The angle of rotation is approximately proportional to the enantiomeric excess of the sample (see box). The problem with this method is that to measure an actual value for ee you need to know what rotation a sample of 100% ee gives, and that is not always possible. Also, polarimeter measurements are notoriously unreliable—they depend on temperature, solvent, and concentration, and are subject to massive error due to small amounts of highly optically active impurities.

Is optical rotation proportional to enantiomeric excess?

Imagine you have a sample, A, of an enantiomerically pure compound—a natural product perhaps—and, using a polarimeter, you find that it has an $[\alpha]_D$ of +10.0. Another sample, B, of the same compound, which you know to be *chemically* pure (perhaps it is a synthetic sample), shows an $[\alpha]_D$ of +8.0. What is its enantiomeric excess? Well, you would have got the same value of 8.0 for the $[\alpha]_D$ of B if you had mixed 80% of your enantiomerically pure sample A with 20% of a racemic (or achiral) compound with no optical rotation. Since you know that sample B is chemically pure, and is the same compound as A, it must therefore indeed consist of 80% enantiomerically pure material plus 20% racemic material, or 80% of one enantiomer plus 20% of a 1:1 mixture of the two enantiomers—which is the same as 90% of one enantiomer and 10% of the other, or 80% enantiomeric excess. Optical rotations can give a guide to enantiomeric excess—sometimes called *optical purity* in this context—but slight impurities of compounds with large rotations can distort the result and there are some examples where the linear relationship between ee and optical rotation fails because of what is known as the Horeau effect. You can read more about this in Eliel and Wilen, *Stereochemistry of organic compounds*, Wiley, 1994.

Chemists now usually use chromatography, or occasionally spectroscopy, to quantify ratios of enantiomers. You may think that this should be impossible—since enantiomers are chemically identical and have identical NMR spectra, how can chromatography or spectroscopy tell them apart? Well, again, they are identical *unless they are in a chiral environment*. We introduced HPLC on a chiral stationary phase as a way of separating enantiomers preparatively in Chapter 14. The same method can be used analytically—less than a milligram of chiral compound can be passed down a narrow column containing silica modified with a chiral additive. One enantiomer passes through the silica faster than the other; the two enantiomers are separated and the quantity of each can be measured (usually by UV absorption or by refractive index changes) and an ee derived. Gas chromatography can be used in the same way—the columns are packed with a chiral stationary phase such as the isoleucine derivative shown in the margin.

➔ This is the principle on which resolution relies: see p. 322.

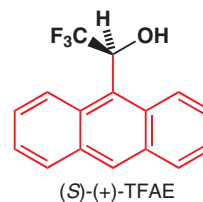


gas chromatography with this chiral stationary phase allows enantiomers to be separated

Distinguishing enantiomers spectroscopically relies again on putting them into a chiral environment. One way of doing this, if the compound is, say, an alcohol or an amine, is to make a derivative (an ester or an amide) with an enantiomerically pure and racemization-proof acyl chloride. The one most commonly used is known as Mosher's acid chloride, after its inventor Harry Mosher, although there are many others. The two enantiomers of the alcohol or amine now become diastereoisomeric esters, and give different sets of peaks in the NMR spectrum—the integrals can be used to determine ee and, although the ^1H NMR of such a mixture of diastereoisomers may become quite cluttered because it is a mixture, the presence of the CF_3 group means that the ratio can alternatively be measured by integrating the two singlets in the otherwise featureless plain of the ^{19}F NMR spectrum.

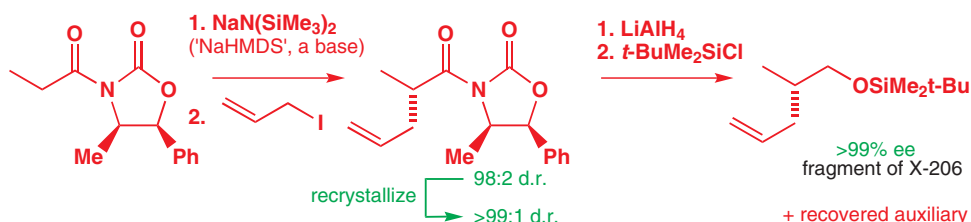


Another powerful method of discriminating between enantiomers is to add an enantiomerically pure compound to the NMR sample that simply forms a complex with the compound under investigation. The complexes formed from the two opposite enantiomers are diastereoisomeric, and therefore have different chemical shifts and, by integrating the NMR signals, the ratio of enantiomers can be determined. Among the most commonly used is this alcohol, 2,2,2-trifluoro-1-(9-anthryl)ethanol, or TFAE, which can both hydrogen-bond to and form



π -stacked complexes with a range of functionalized compounds, and often splits NMR signals due to enantiomeric compounds very cleanly.

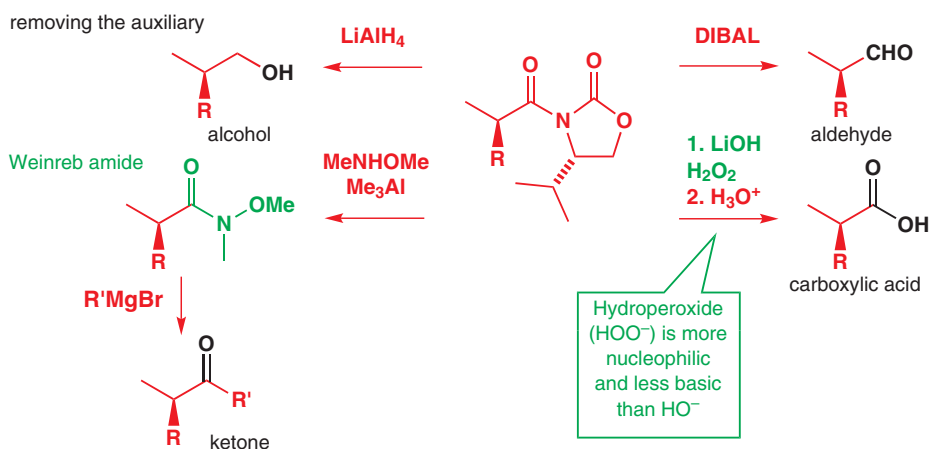
Time to go back to chiral auxiliaries. We pointed out that, although we want to get maximum levels of stereoselectivity in our chiral-auxiliary-controlled reaction, we may still have a small percentage of a minor diastereoisomer, which, once we have removed our chiral auxiliary, will compromise the ee of our final product. It is at this point that we can use a trick that essentially employs the chiral auxiliary in a secondary role as a resolving agent. Provided the products are crystalline, it will usually be possible to recrystallize our 94:6 mixture of diastereoisomers to give essentially a single diastereoisomer, rather like carrying out a resolution with an enormous head start. Once this has been done, the chiral auxiliary can be removed and the product may be very close to 100% ee. Of course, the recrystallization sacrifices a few percentage points of yield, but these are invariably much less valuable than the few percentage points of ee gained! Here is an example from the work of Evans himself. During his synthesis of the complex antibiotic X-206 he needed large quantities of the small molecule below. He decided to make it by a chiral-auxiliary-controlled allylation, followed by reduction to give the alcohol. The auxiliary needed is the one derived from norephedrine, and the reaction of the enolate with allyl iodide gives a 98:2 mixture of diastereoisomers. However, recrystallization converts this into an 83% yield of a single diastereoisomer in >99% purity, giving material of essentially 100% ee after removal of the auxiliary.



■ At this point we should also come clean about the asymmetric Diels–Alder reaction we introduced on p. 1108: it is not quite as selective as we implied—a minor diastereoisomer is formed in a 7% yield, with the major isomer accounting for 93%. But just one recrystallization gives >99% diastereoisomerically pure material in 81% yield.

This is one big bonus of using a chiral auxiliary—it's much easier to purify diastereoisomers than enantiomers and a chiral auxiliary-controlled reaction necessarily produces diastereoisomeric products.

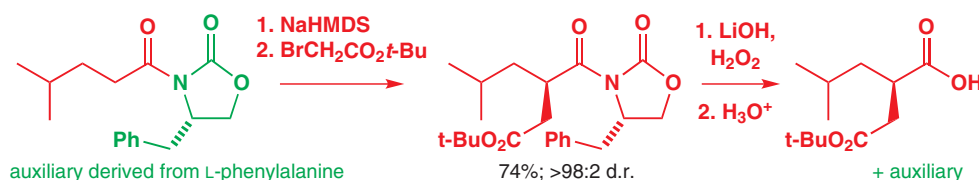
Both these examples of auxiliary-controlled alkylation make use of LiAlH_4 reduction to the alcohol in the step which removes the auxiliary. You saw attack with an alkoxide above, and several other alternative methods are possible as well, summarized below. DIBAL ($i\text{-Bu}_2\text{AlH}$, p. 533) reduces the product to an aldehyde, while converting the product to a Weinreb amide (p. 219) makes formation of a ketone possible.



Epimers are pairs of diastereoisomers differing in configuration at just one chiral centre.

Epimerization is the interconversion of such diastereoisomers just as racemization is the interconversion of enantiomers.

Simple hydrolysis under acid or basic conditions risks epimerizing the newly created chiral centre, and a good solution is to use the less basic, more nucleophilic hydroperoxide anion. This was the approach taken by chemists making this component of a collagenase inhibitor. Notice that this auxiliary is a variant based on L-phenylalanine.

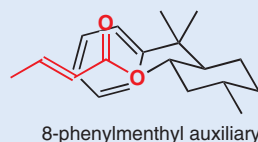
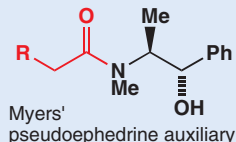
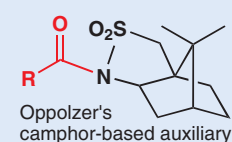


These various ways of removing auxiliaries illustrate the ways in which it is possible to make a virtue out of one of their big disadvantages: chiral auxiliaries must first be attached to the compound under construction, and after they have done their job they must be removed. The best auxiliaries can be recycled, but even then there are still at least two 'unproductive' steps in the synthesis.

➡ The reason for the greater nucleophilicity of the hydroperoxide anion is discussed in Chapter 22, p. 513.

Oxazolidinones are not the only auxiliaries

Other auxiliaries are also used, and the choice of auxiliary may depend not only on the selectivity of the reaction under investigation but also on the physical properties of the products. The camphor-based auxiliary of Oppolzer is reputed to confer crystallinity on its derivatives, while the pseudoephedrine auxiliary of Myers is cheap, readily available, and very easy to introduce. More bulky auxiliaries such as 8-phenylmenthol work well where control over long-range interactions, such as conjugate additions, are required.



🔍 Interactive mechanism for Oppolzer's sultam in conjugate addition

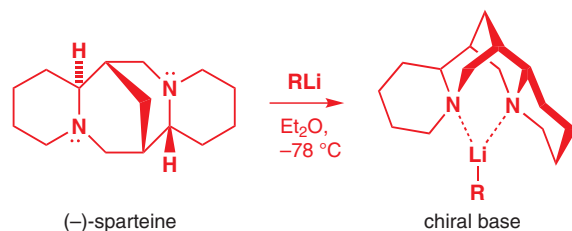
🔍 Interactive mechanism for 8-phenylmenthol in Diels–Alder reaction

Chiral reagents

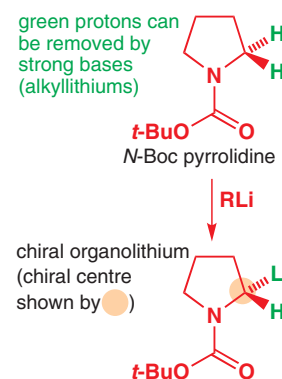
A chiral auxiliary is a chiral molecule attached to the starting material of the reaction; diastereoselective reactions of compounds from the chiral pool are likewise controlled by chirality in the starting material, and we call this type of stereocontrol *substrate control*. But it is also possible for enantioselective reactions to be controlled by chiral *reagents*. For example, a typical achiral base will just remove a proton from a substrate, but an enantiomerically pure chiral base can select one of two enantiotopic protons and form a product enantioselectively. The product of course has to be chiral, so we can't use a chiral base to make planar enolates enantioselectively, for example, but we can a chiral base to make chiral organolithiums.

Alkylolithiums are sufficiently strong as bases to remove the protons adjacent to the nitrogen atom of *N*-Boc pyrrolidine, shown in the margin. The product of deprotonation is an organolithium which is a chiral molecule: the lithium-bearing carbon is chiral.

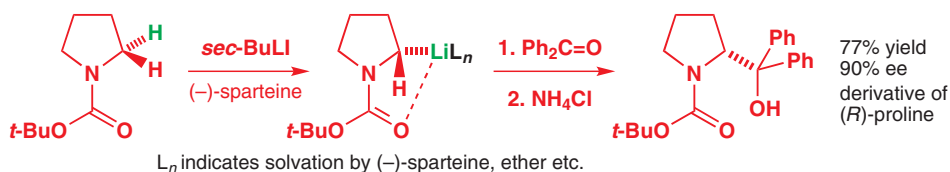
Alkylolithiums can be turned into chiral bases in quite a simple way—by complexation with a chiral ligand. A widely used example is the tetracyclic diamine (–)-sparteine. Sparteine's structure looks complex, but it is a relatively widely available natural product which folds around the lithium atom of an alkylolithium and places the base in a chiral environment.



This chiral base can now choose to remove from the pyrrolidine substrate just one of the enantiotopic protons adjacent to nitrogen, and form a chiral, enantiomerically enriched organolithium. The stereochemistry of the organolithium is preserved through its reactions with electrophiles such as the ketone shown here.



Interactive mechanism for sparteine-mediated lithiation



One of the reasons this reaction is so useful is that the products happen to be derivatives of the less readily available (*R*)-proline. But, as with chiral auxiliaries, if you use a chiral reagent you need a full equivalent of the source of enantiomeric purity (here, (–)-sparteine) which can get prohibitively expensive on a large scale. It is for this reason that the real pinnacles of achievement in asymmetric synthesis make use of *asymmetric catalysis*, which we turn to next.

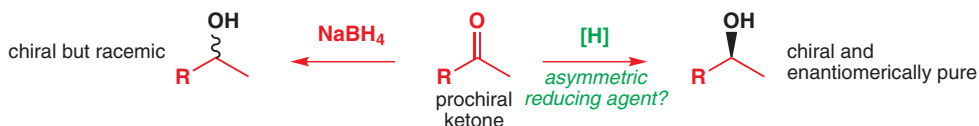
Asymmetric catalysis

➡ Look back at Chapters 31 and 33 (p. 820) if you need reminding about the terms prochiral, enantiotopic, and diastereotopic.

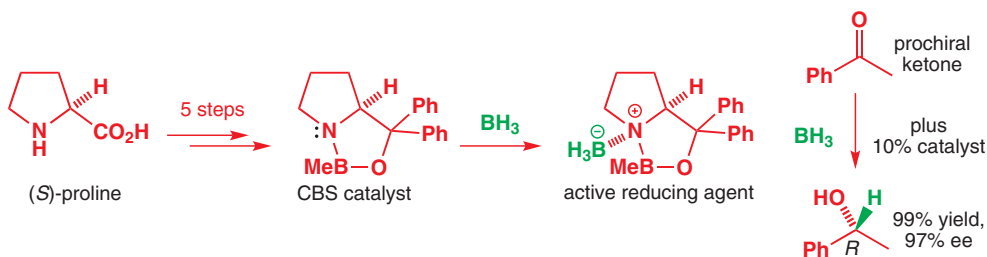
If we want to create a new chiral centre in a molecule, our starting material must have **prochirality**—the ability to become chiral in one simple transformation. The most common prochiral units that give rise to new chiral centres are the trigonal carbon atoms of alkenes and carbonyl groups, which become tetrahedral by addition reactions. In the last section you saw a prochiral, tetrahedral CH_2 group becoming a chiral organolithium by enantioselective removal of one enantiotopic proton. Much more common are the reactions you saw in the section before that, where in every case a prochiral alkene (we can count enolates as alkenes for this purpose) reacted selectively on one face because of the influence of the chiral auxiliary, which made the faces of the alkene diastereotopic.

Catalytic asymmetric reduction of ketones

One of the simplest transformations you could imagine of a prochiral unit into a chiral one is the reduction of a ketone. Although chiral auxiliary strategies have been used to make this type of reaction asymmetric, conceptually the simplest way of getting the product as a single enantiomer would be to use a chiral reducing agent—in other words, to attach the chiral influence not to the *substrate* (as we did with chiral auxiliaries) but to the *reagent*. We need an asymmetric version of NaBH_4 .



One of the more widely used solutions to this challenge is the chiral borohydride analogue invented by Itsuno in Japan and developed by Corey, Bakshi, and Shibata. It is based on a stable boron heterocycle made from an amino alcohol derived from proline (see the box below for the synthesis), and is known as the CBS catalyst after its developers. The active reducing agent is generated when the heterocycle forms a complex with borane. Only catalytic amounts (usually about 10%) of the boron heterocycle are needed because borane is sufficiently reactive to reduce ketones only when complexed with the nitrogen atom. The rest of the borane just waits until a molecule of catalyst becomes free.

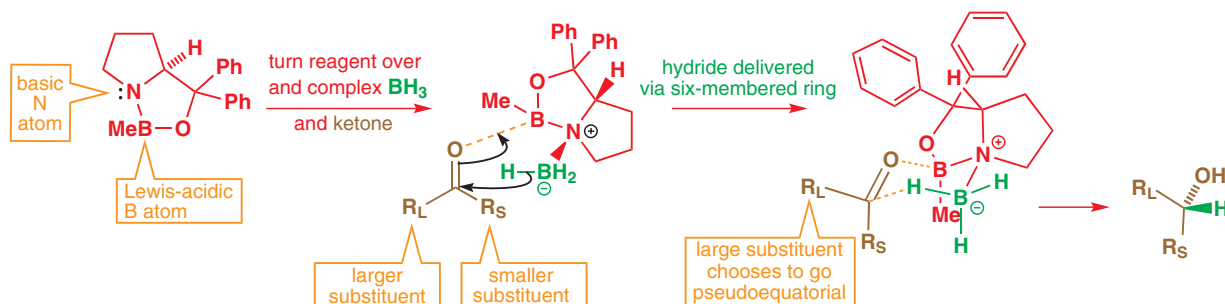


■ The fact that the reactions are catalytic in the heterocycle means that relatively little is needed. Note the distinction from chiral auxiliaries here: although auxiliaries are recoverable, they always have to be used in stoichiometric quantities and recovery is usually a separate step. Later in the chapter you will see catalytic reactions that use 1000 times less catalyst than this one.

CBS reductions are best when the ketone's two substituents are well-differentiated sterically—just as Ph and Me are in the example above. The reaction works because the heterocyclic

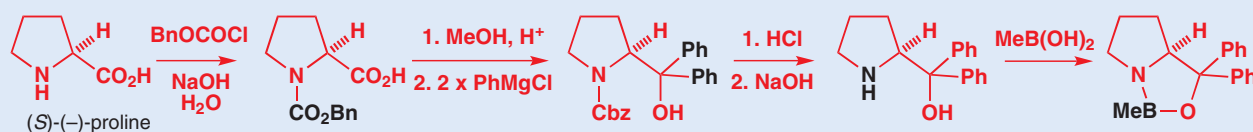
catalyst brings together the borane (which complexes to its basic nitrogen atom) and the carbonyl compound (which complexes to its Lewis-acidic boron atom). Complexation activates both partners towards reaction: donating electron density to the borane is essential to persuade it to transfer hydride, and withdrawing electron density from the carbonyl group makes it electrophilic enough to react with a weak hydride source. The hydride is delivered via a six-membered cyclic transition state, with the enantioselectivity arising from the preference of the larger of the ketone's two substituents (R_L) for the pseudoequatorial position on this ring.

 Interactive asymmetric reduction of ketone with CBS catalyst



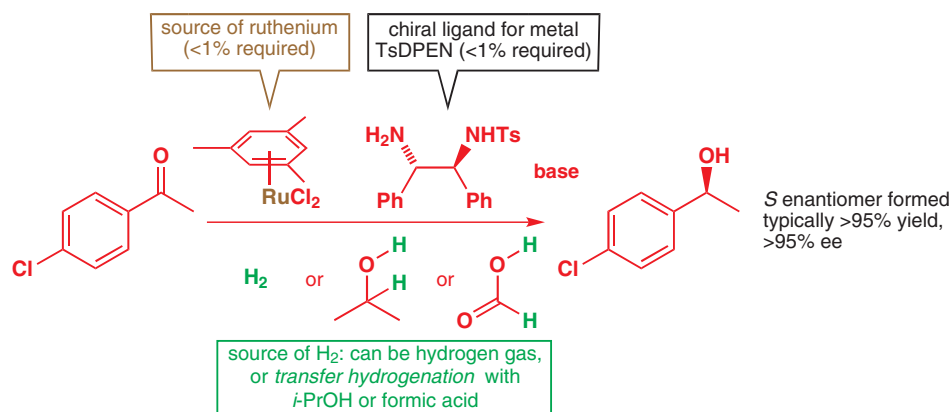
Making the CBS catalyst

To make the CBS heterocycle, (*S*)-proline has to be protected as its *N*-Cbz derivative (Chapter 23) and converted to its methyl ester. Esters react with Grignard twice to give tertiary alcohols (Chapter 10), so PhMgBr followed by deprotection gives the amino alcohol needed. Condensation with methylboronic acid (MeB(OH)₂) gives the stable catalyst.



To make the other enantiomer, you would need the rather more expensive 'unnatural' (*R*)-proline, which you can make by the method of p. 1114, but in such a case you might consider using one of the alternative reduction methods described below.

Until recently, the CBS reagent was one of the most commonly used asymmetric reducing agents for ketones. But in the early years of the 21st century a new reaction has taken over that role—one in which the job of bringing together the ketone and the reducing agent is taken by an atom of ruthenium. The ruthenium is added as Ru(II) in a 16-electron complex (see p. 1116) with an aromatic compound such as 1,3,5-trimethylbenzene (known as mesitylene). A chiral ligand is needed—the diamine derivative shown here is best. Only very small amounts (often << 1%) of the catalyst and ligand are required, which is a good thing as both are much more expensive than the reagents in the CBS reduction. The reducing agent itself can be hydrogen or, more conveniently, a more easily handled source of hydrogen atoms such as isopropanol (which gets oxidized to acetone) or formic acid (which gets oxidized to carbon dioxide). Here's a typical example; we will explain how it works shortly.



This revolution in asymmetric catalysis using chiral complexes of transition metals was made possible principally by the work of Ryoji Noyori (who developed the Ru- and Rh-catalysed reductions we describe in this chapter) and of K. Barry Sharpless (who developed the Os- and Ti-catalysed oxidations). This work won Noyori and Sharpless the Nobel Prize for Chemistry in 2001, along with William Knowles (who was the first to apply metal-catalysed asymmetric reactions to industrial targets).

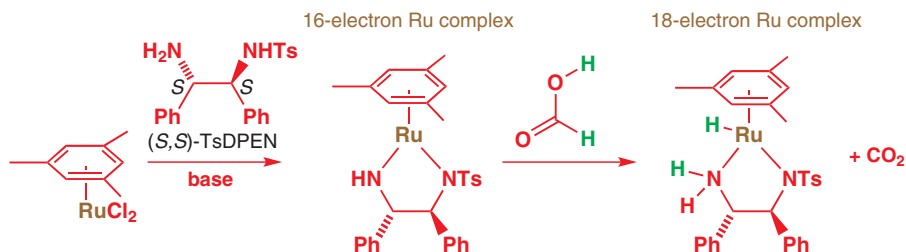
■ This section discusses several reactions of organometallic compounds. In order to understand their mechanisms you must be familiar with the terminology associated with organometallic complexes, such as how to 'count electrons', described in Chapter 40.

Interactive mechanism for ruthenium-catalysed ketone hydrogenation

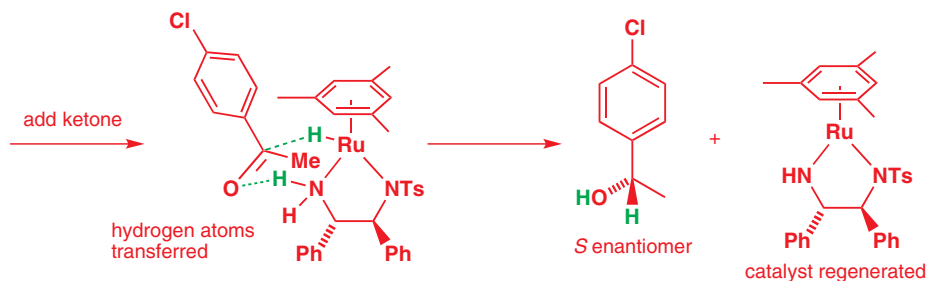
■ To show how fast this area is moving, we can quote from the first edition of this textbook, published in 2001: 'You would not normally choose catalytic hydrogenation for reducing a carbonyl group to an alcohol and, indeed, carbonyl reductions using hydrogenation with a chiral catalyst are not usually very enantioselective.' A lot has changed in little more than a decade.

You have met several reactions of ruthenium complexes, especially in Chapters 38 and 40, where you saw ruthenium carbenes catalysing the metathesis of alkenes. Ruthenium is one of a select group of transition metals (Pd, Ru, Rh, Cu, Os, and Ti being the others) which play an important role in asymmetric catalysis. The key to their success is the transition metal coordination chemistry we looked at in the last chapter: the metals can act as coordination sites for substrates, and by using other ligands which are chiral and enantiomerically pure, the reactions they catalyse can be made to take place in an asymmetric environment.

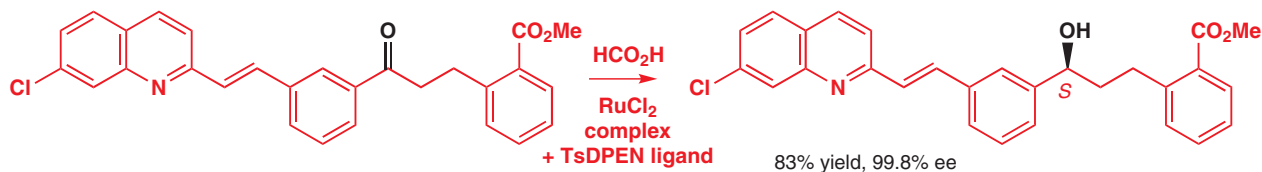
The ruthenium-catalysed reduction of ketones starts with coordination of the tosyl-diamine ligand ((*S,S*)-*N*-toluenesulfonyl 1,2-phenylenediamine, or 'TsDPEN') to the ruthenium metal. This is a 16-electron complex, and can be reduced by formic acid to an 18-electron ruthenium hydride.



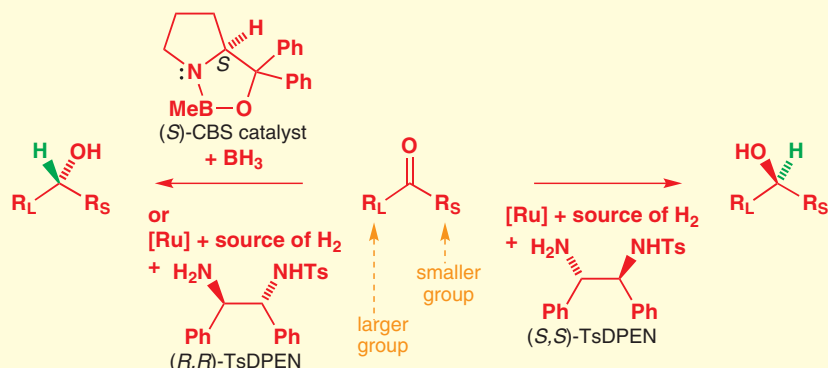
Now comes the reduction. Provided the ketone approaches the ruthenium complex in the right orientation, with the smaller methyl group tucked in under the ruthenium and the larger aryl group pointing away from the bulky ligands, the 18e complex can transfer the carbonyl group simultaneously H⁻ from Ru and H⁺ from the protonated nitrogen. The chiral ligand means that the alcohol is formed as a single enantiomer, and the ruthenium catalyst is regenerated.



The reduction shown below is particularly important because it generates a late intermediate in the industrial synthesis of the anti-asthma drug montelukast (Singulair). Several methods have been used, but in 2008 chemists at the Croatian pharmaceutical company Pliva patented a method using the ruthenium catalyst with a derivative of TsDPEN as a ligand to give the product in 83% yield and 99.8% ee on a scale of several kilograms.

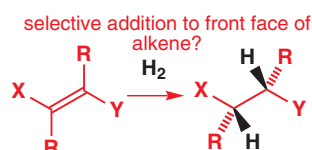
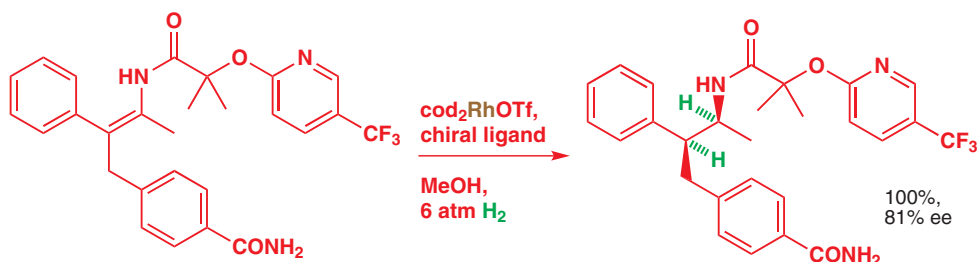


● Two methods for reducing carbonyl compounds enantioselectively



Catalytic asymmetric hydrogenation of alkenes

Reduction of a ketone can give a chiral secondary alcohol, but reduction of an alkene by addition of hydrogen to one of its two enantiotopic faces can give all sorts of products, creating either one or two chiral centres, depending on the substituents on the alkene. By way of illustration (and explanation will follow soon) the alkene hydrogenation below creates, in one step, the two chiral centres of a precursor to the anti-obesity drug taranabant. A single diastereoisomer is formed by *syn* addition of hydrogen, and a chiral ligand ensures that one enantiomer is formed preferentially.

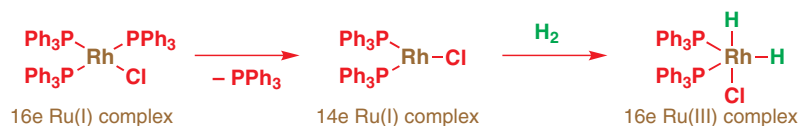
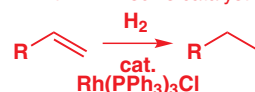


■ cod is the ligand cyclooctadiene, and 6 atmospheres pressure of H_2 is used.

You have seen numerous hydrogenations of alkenes using hydrogen over a solid catalyst of palladium supported on charcoal ('heterogeneous hydrogenation'), but catalytic asymmetric hydrogenation of alkenes uses a different type of catalyst—a soluble complex, often of Ru or Rh with phosphine-containing ligands. The substrates for asymmetric alkene hydrogenation are also more limited than those for hydrogenation with Pd/C because they must carry a functional group close to the alkene, allowing coordination to the transition metal catalyst. In the example above, that functional group is the amide directly adjacent to the alkene.

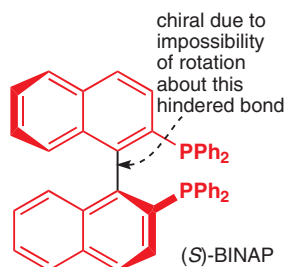
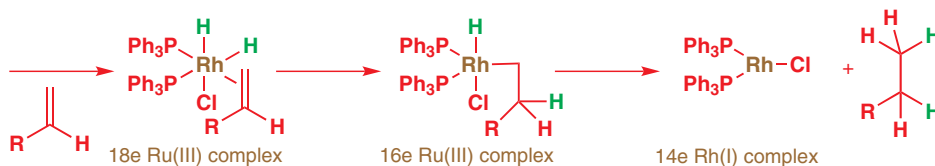
The inspiration for these catalysts came from the work of Wilkinson in London in the 1960s, who showed that $RhCl(PPh_3)_3$ (known as Wilkinson's catalyst) promoted *homogeneous* hydrogenation of alkenes (i.e. during the reaction there is only one, solution, phase). Wilkinson's catalyst is a 16-electron complex of Rh(I), and it works as a catalyst because it can easily lose one of the phosphine ligands to form a 14-electron complex. This undergoes addition of H_2 , giving a 16-electron Ru(III) complex.

homogeneous hydrogenation with Wilkinson's catalyst



➔ See p. 1074 for more on Wilkinson's catalyst.

The complex is still coordinatively unsaturated, so an alkene can form a π complex with a full complement of 18 electrons. Migratory insertion of one of the hydrogen atoms, followed by reductive elimination, gives back the 14-electron Rh complex and the reduced alkane.

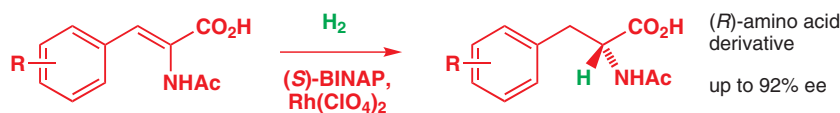


■ The detailed mechanism of this reaction is too complicated for us to examine here. Unusually, it involves two diastereoisomeric complexes, of which the more reactive is the one which is less favoured.

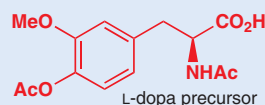
The conceptual advance which allows this sort of hydrogenation to become asymmetric is the replacement of the two achiral triphenylphosphine ligands of Wilkinson's catalyst with chiral phosphine ligands. Notice that two of the triphenylphosphines remain coordinated through the whole reaction mechanism, so by doing this we can ensure that the Rh always remains in a chiral environment.

The usual solution is to use one chiral molecule containing two phosphorus atoms, and the most important of these is BINAP. BINAP is a chelating diphosphine: the metal sits between the two phosphorus atoms, firmly anchored in a chiral environment. The chirality here is of an unusual sort, since BINAP, which you met in Chapter 14 (p. 319), has no chiral centres. It is one of the class of chiral molecules, known as *atropisomers*, whose chirality arises from the inability of the bond between the two naphthalene rings to rotate.

Incorporating (S)- or (R)-BINAP into a hydrogenation with Rh can lead to high enantiomeric excesses in the products because during the migratory insertion step the complex is forced to transfer hydrogen to only one of the two possible enantiotopic faces of the alkene. As we remarked before, asymmetric hydrogenations require a functional group which can coordinate to the metal, and with Rh the best substrates are *N*-acyl enamines. Those of the type shown below give excellent results, and usefully with (S)-BINAP the products are protected amino acids of the opposite enantiomeric series to those found in nature.



Industrial synthesis of L-dopa

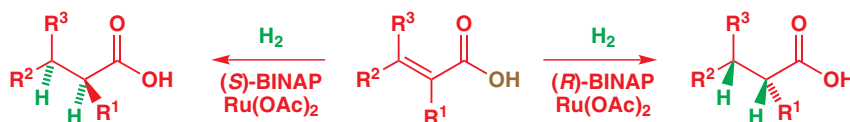


A related hydrogenation to give the product above, using a different catalyst, was developed by William Knowles at Monsanto. This hydrogenation was the first demonstration of the use of asymmetric catalysis in the synthesis of a chiral drug, and Knowles shared the Nobel Prize in 2001 with Noyori and Sharpless.

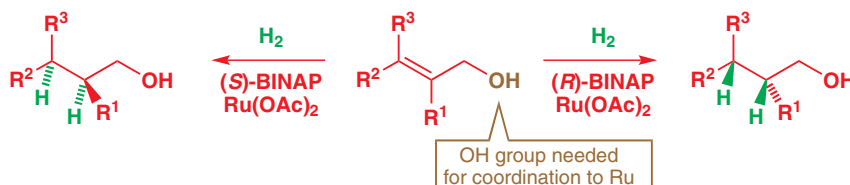
It is even economical for the more expensive of the natural amino acids to be made synthetically by this type of reaction rather than isolated from natural sources—phenylalanine, which is of industrial importance as a component of the artificial sweetener aspartame (NutraSweet), is manufactured using enantioselective hydrogenation.

Noyori found that using ruthenium instead of rhodium broadens greatly the scope of the substrates that will undergo asymmetric hydrogenation. They still need a functional group—usually the OH group of an alcohol or a carboxylic acid—to allow coordination to the metal, but the alkene itself can be a simple allylic alcohol or unsaturated acid derivative. BINAP is again a good ligand, and of course by choosing which enantiomer of BINAP you use you can choose which enantiomer of the product you get.

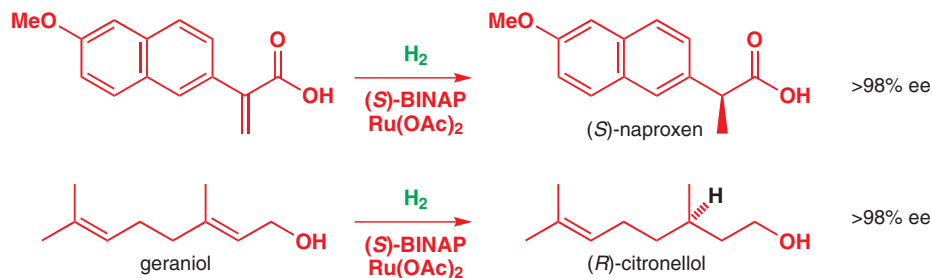
Ru-catalysed asymmetric hydrogenation of unsaturated carboxylic acids



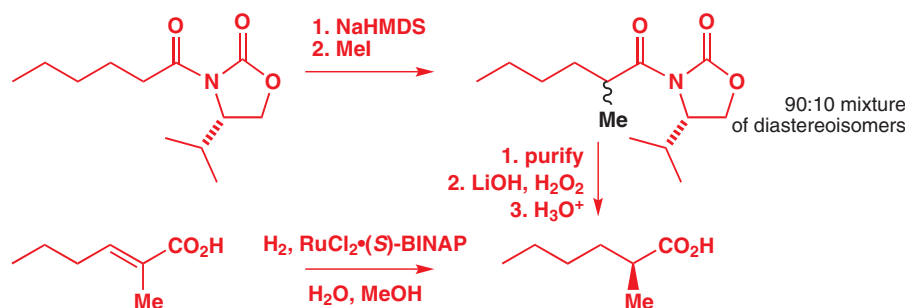
Ru-catalysed asymmetric hydrogenation of allylic alcohols



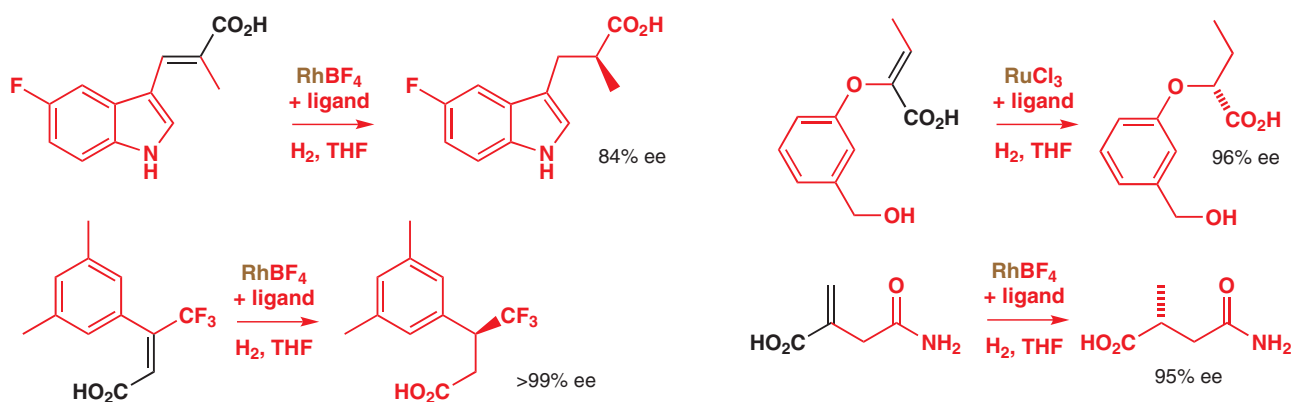
Two important industrial asymmetric syntheses which routinely use this chemistry are the production of the painkiller (*S*)-naproxen and the synthetic intermediate and perfumery compound (*R*)-citronellol. It is gratifying to note that this chemistry, using <1% of Ru, gives citronellol in higher enantiomeric purity than many natural sources of the same compound!



Reduction of unsaturated carboxylic acids gives products that you might alternatively think of making by auxiliary-controlled alkylation methods. When the NutraSweet company needed this chiral branched carboxylic acid as a single enantiomer, they initially used the auxiliary methods of p. 1110 to make a small amount, but they found that ruthenium-catalysed hydrogenation was greatly to be preferred on a large scale: just 22 g of the ruthenium- (*S*)-BINAP complex is needed to produce 50 kg of product with 90% ee.

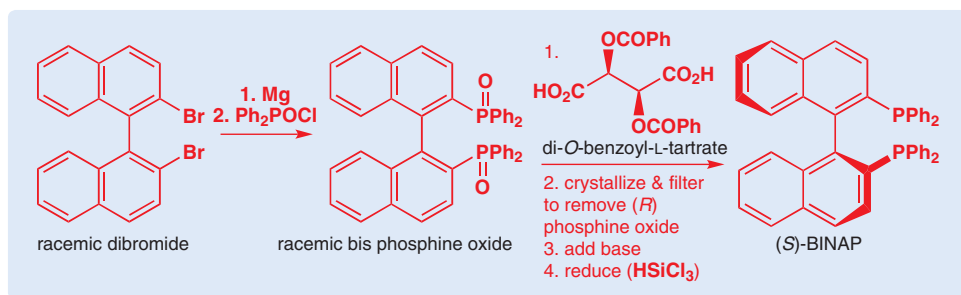


In the last 20 years, the variety of ligands available for rhodium and ruthenium-catalysed hydrogenations has increased to the point where the right combination of metal and ligand will reduce almost any unsaturated carboxylic acid derivative in high enantiomeric excess. Details are beyond the scope of this book, but we leave you with four examples, all from industrial drug syntheses, to illustrate how versatile the method can be.



Resolution of BINAP

BINAP is not derived from a natural product, and has to be synthesized in the laboratory and resolved using a naturally derived resolving agent. The scheme shows one method by which enantiomerically pure BINAP may be made—the resolution step is unusual because it relies on formation of a molecular complex, not a salt. The bis phosphine oxide of (*S*)-BINAP co-crystallizes with di-*O*-benzoyl-L-tartrate, leaving the (*R*)-phosphine oxide in solution. Base releases the pure (*R*)-phosphine oxide that is resolved, which is then reduced to the phosphine with trichlorosilane.

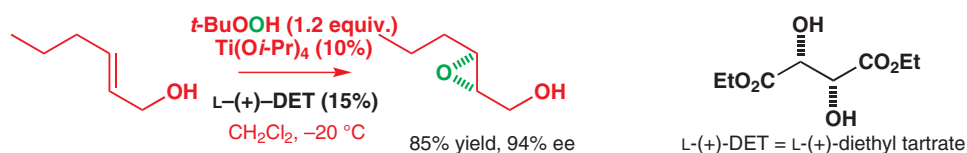


Asymmetric epoxidation

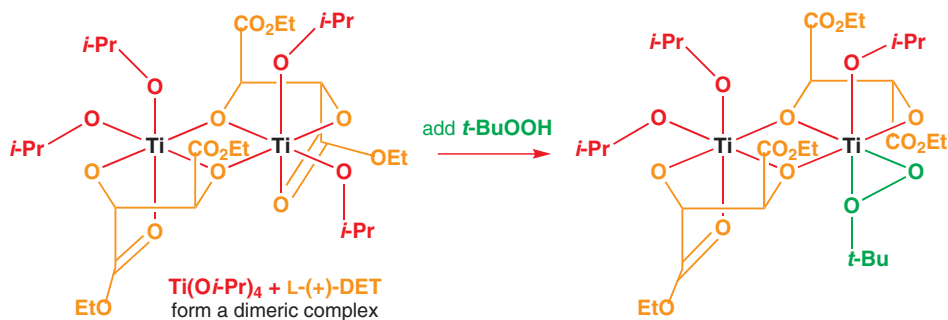
K. B. Sharpless (1941–) studied at Stanford and was first appointed at MIT but is now at the Scripps Institute in California. His undoubted claim to fame rests on the invention of no fewer than three reactions of immense significance: asymmetric epoxidation (AE) and asymmetric dihydroxylation (AD) are discussed in this chapter. The third reaction, asymmetric aminohydroxylation (AA), has still to reach the perfection of the first two.

Asymmetric *hydrogenation* of an alkene can create two new chiral centres, but introduces no new functionality as it does so. Asymmetric *oxidation* of an alkene is different: it can create two new chiral centres and two new functional groups at the same time. We will now look at two examples of asymmetric oxidation, both products of the laboratories of Professor Barry Sharpless.

The first of Sharpless's reactions is an oxidation of alkenes by asymmetric epoxidation. You met vanadium as a transition-metal catalyst for epoxidation with *t*-butyl hydroperoxide in Chapter 32, and this new reaction makes use of titanium, as titanium tetraisopropoxide, $\text{Ti}(\text{O}i\text{-Pr})_4$, to do the same thing. Sharpless and his co-worker Tsutomu Katsuki surmised that by adding a chiral ligand to the titanium catalyst they might be able to make the reaction asymmetric. The ligand that works best is diethyl tartrate, and one example of the reaction is shown below.

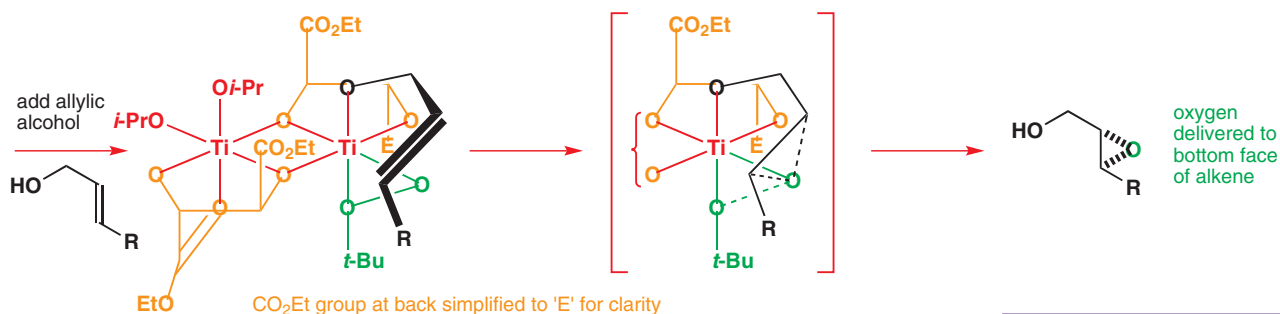


Transition-metal-catalysed epoxidations work only on allylic alcohols, so this is a limitation of the method, but otherwise there are few restrictions on what can be epoxidized enantioselectively, and when this reaction was discovered in 1981 it was by far the best asymmetric reaction known. Because of its importance, a lot of work went into discovering exactly how the reaction worked, and the scheme below shows what is believed to be the active complex, formed from two titanium atoms bridged by two tartrate ligands (shown in orange). Each titanium atom retains two of its isopropoxide ligands and is coordinated to one of the carbonyl groups of the tartrate ligand. The reaction works best if the titanium and tartrate are left to stir for a while so that these dimers can form cleanly. When the oxidizing agent (*t*-BuOOH, shown in green) is added to the mixture, it displaces one of the remaining isopropoxide ligands and one of the tartrate carbonyl groups.



For this oxidizing complex to react with an allylic alcohol, the alcohol must become coordinated to the titanium too, displacing a further isopropoxide ligand. Because of the shape

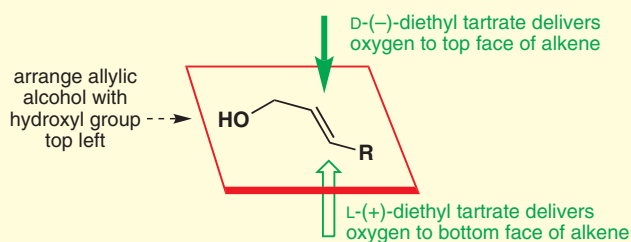
of the complex the reactive oxygen atom of the bound hydroperoxide has to be delivered to the lower face of the alkene (as drawn), and the epoxide is formed in high enantiomeric excess. Displacement of the product by another molecule of hydroperoxide starts the cycle again.



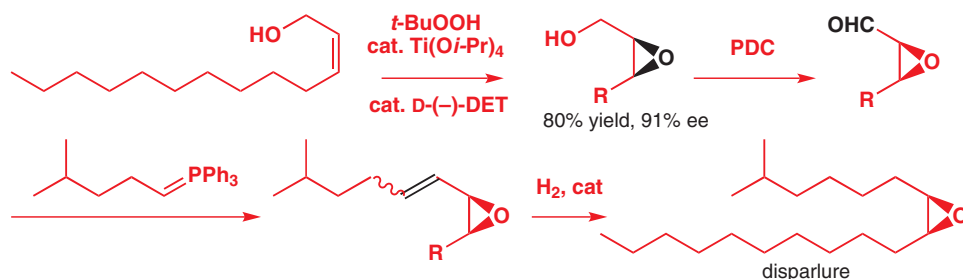
Different allylic alcohols coordinate in the same way to the titanium and reliably present the same enantiotopic face to the bound oxidizing agent, and the preference for oxidation with L-(+)-DET is shown in the schematic diagram below. Tartrate is ideal as a chiral ligand because it is available relatively cheaply as either enantiomer. L-tartrate is extracted from grapes; D-(–)-tartrate is rarer and more expensive, but still cheap by the standards of some of the bisphosphine ligands used in the last section. By using D-(–)-tartrate it is, of course, possible to produce the other enantiomer of the epoxide equally selectively.

 Interactive mechanism for the Sharpless epoxidation of allylic alcohols

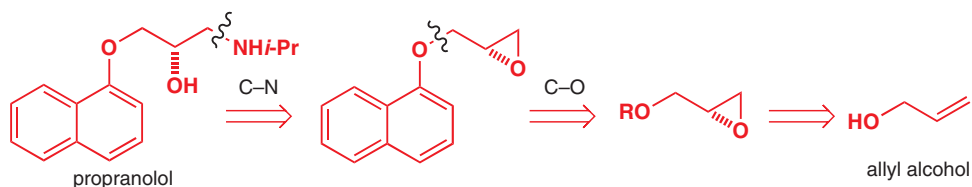
● Enantioselectivity in the Sharpless asymmetric epoxidation



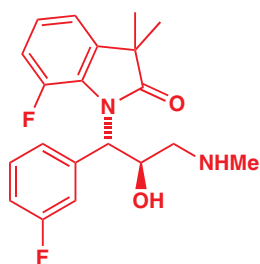
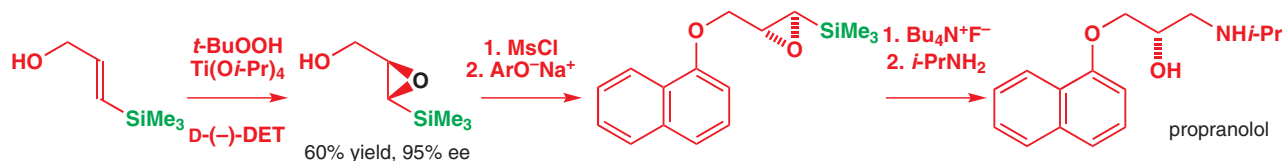
Sharpless also found that this reaction works with only a catalytic amount of titanium-tartrate complex because the reaction products can be displaced from the metal centre by more of the two reagents. The catalytic version of the asymmetric epoxidation is well suited to industrial exploitation, and the American company J. T. Baker has employed it to make synthetic disparlure, the pheromone of the gypsy moth, by oxidation of the epoxy alcohol to an aldehyde with pyridinium dichromate (PDC) (p. 543), Wittig reaction (p. 689), and hydrogenation.



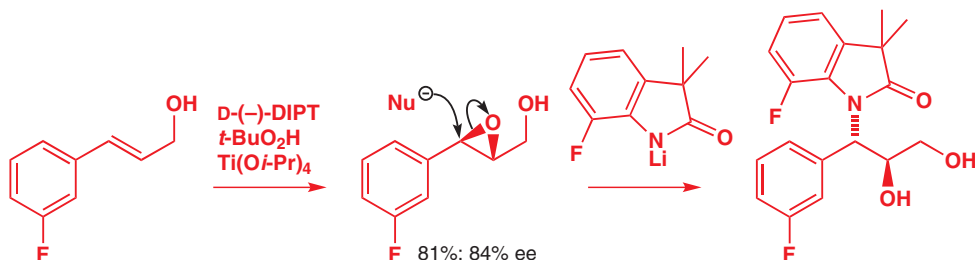
Not many target molecules are themselves epoxides, but the great thing about the epoxide products is that they are highly versatile—they react with many types of nucleophiles to give 1,2-disubstituted products. You met the chiral beta-blocker drug propranolol in Chapter 28, and its 1,2,3-substitution pattern makes it a good candidate for synthesis using asymmetric epoxidation.



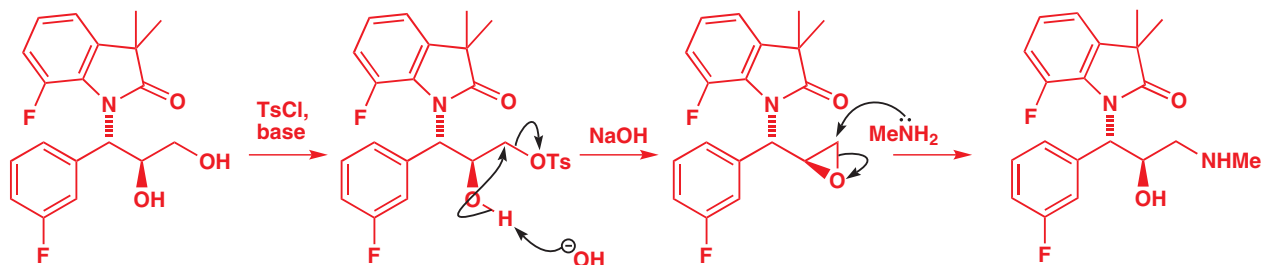
Unfortunately, the obvious starting material, allyl alcohol itself, gives an epoxide that is hard to handle, so Sharpless, who carried out this synthesis of propranolol, used this silicon-substituted allylic alcohol instead. The hydroxyl group was mesylated and displaced with 1-naphthoxide and, after treatment with fluoride to remove the silicon, the epoxide was opened with isopropylamine.



Chemists at the drug company Wyeth needed the amine shown in the margin. The 1,2,3-functional group pattern led them to think of using the Sharpless asymmetric epoxidation, and epoxidation of a fluorinated allylic alcohol using D-(-)-diisopropyl tartrate (DIPT) gave them the enantiomer they wanted with slightly better selectivity than diethyl tartrate. The benzylic end of the epoxide is more reactive towards nucleophilic substitution, and in the presence of $\text{Ti}(\text{O}i\text{-Pr})_4$, this time simply acting as a Lewis acid, the lithiated heterocycle opens the epoxide with inversion of configuration.

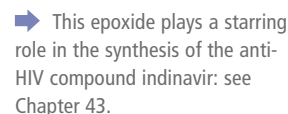
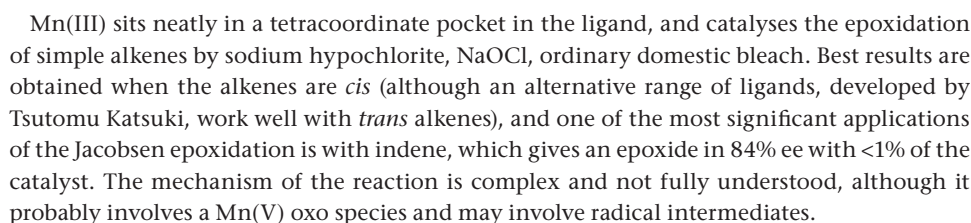


Finally, it's necessary to bring in the amino group, and this can be done by tosylating the less hindered primary hydroxyl group selectively, closing to an epoxide in base, and then reopening the epoxide at the less hindered terminal position with methylamine.



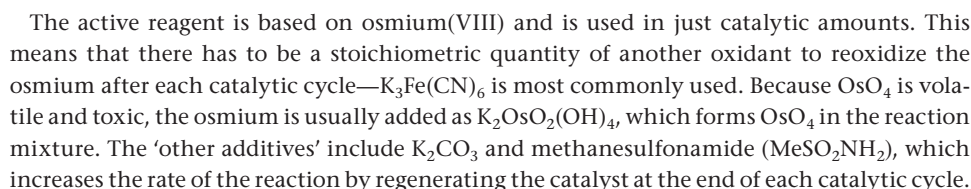
■ 'Salen' is an abbreviation of salicylthylenediamine and simpler salens had long been used as tetradentate ligands for coordination chemistry.

The Sharpless asymmetric epoxidation is reliable, but it works only for allylic alcohols. There is an alternative, however, which works with simple alkenes. The method was developed by Eric Jacobsen and employs a manganese catalyst with a chiral ligand built from a simple diamine. The diamine is not a natural compound and has to be made in enantiomeric form by resolution, but at least that means that both enantiomers are readily available. The diamine is condensed with a derivative of salicylaldehyde to make a bis-imine known as a 'salen'.

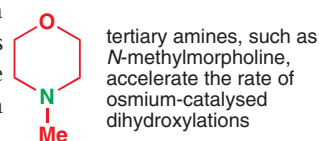


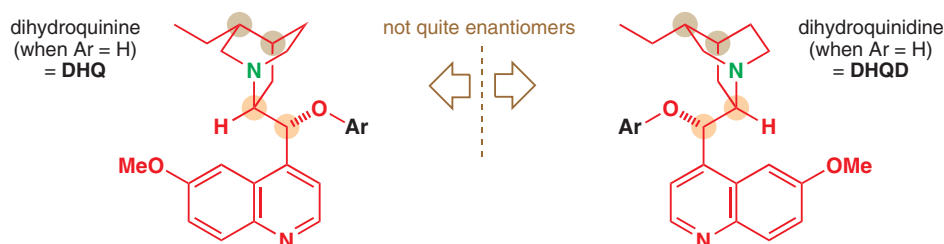
Asymmetric dihydroxylation

asymmetric dihydroxylation—the reaction:

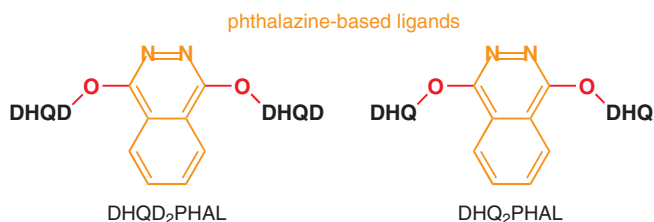


Now for the chiral ligand. Tertiary amines are good ligands for osmium and increase the rate of dihydroxylations: one of the reasons that NMO is used in the racemic version of the reaction (see p. 442) is that the by-product, *N*-methylmorpholine, accelerates the reaction. Sharpless chose some available chiral tertiary amines as ligands, and it turned out that the best ones are based on the alkaloids dihydroquinidine and dihydroquinine, whose structures are shown below. They coordinate to the osmium through the green nitrogen atom.





The alkaloids (usually abbreviated to DHQD and DHQ, respectively) must be attached to an aromatic group Ar, the choice of which varies according to the substrate. The most generally applicable ligands are these two phthalazines in which each aromatic group Ar carries two alkaloid ligands, either DHQ or DHQD.

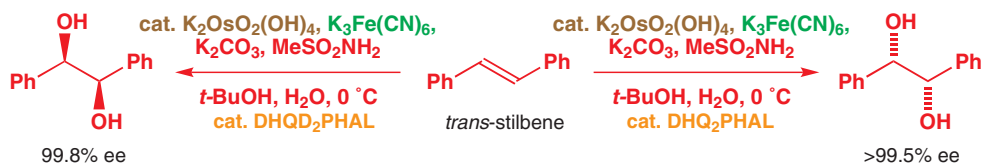


Dihydroquinine and dihydroquinidine are not enantiomeric (although the chiral centres ringed in orange are of opposite configuration in each of the pairs, those ringed in brown remain the same in both), but they act on the dihydroxylation as though they were.

Here, after all that introduction, is a real example, and probably the most remarkable of any in this chapter. *trans*-Stilbene dihydroxylates more selectively than any other alkene, and this particular example is one of the most enantioselective catalytic reactions ever invented.

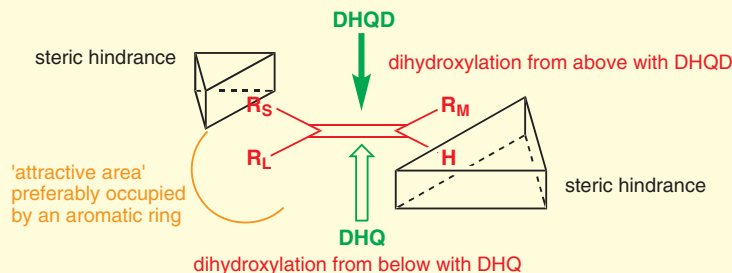
AD-mix

A mixture of all the reagents except MeSO₂NH₂ can be purchased commercially as AD-mix. AD-mix- α contains DHQ₂PHAL, while AD-mix- β contains DHQD₂PHAL.



We can sum up the usual selectivity of the AD reaction with the diagram shown below. With the substrate arranged as shown, with the largest (R_L) and next largest (R_M) groups bottom left and top right, respectively, DHQD-based ligands will direct OsO₄ to dihydroxylate from the top face of the alkene and DHQ-based ligands the bottom face.

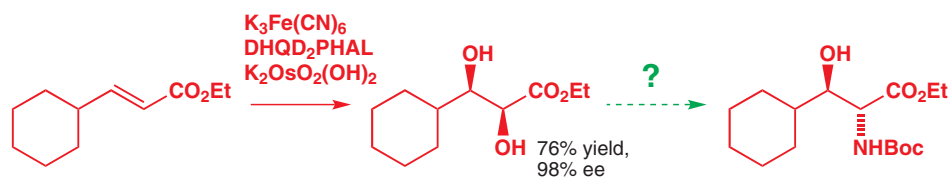
• Enantioselectivity in the Sharpless asymmetric dihydroxylation



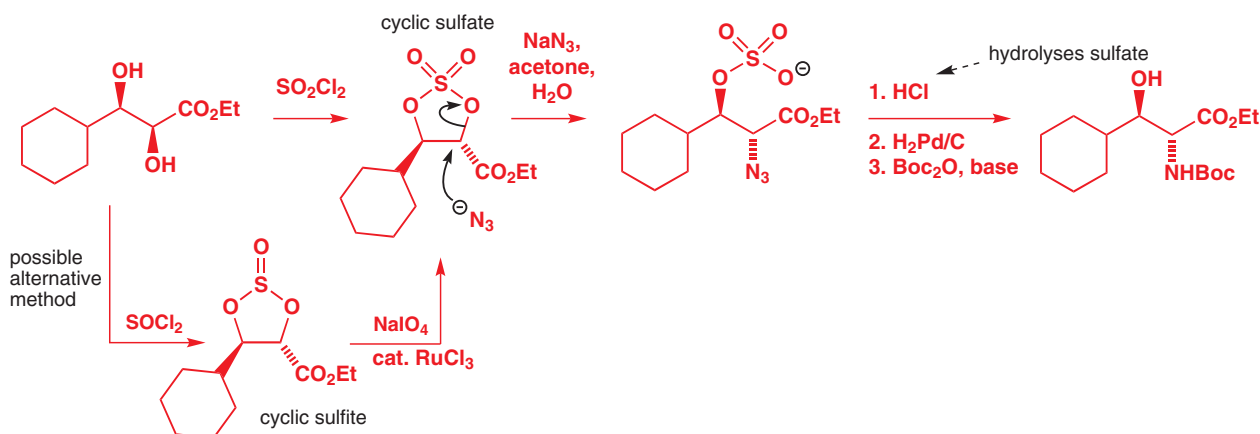
The reason for this must come from the way in which the substrate interacts with the osmium–ligand complex. However, the detailed mechanism of the asymmetric dihydroxylation is still far from clear-cut. What is known is that the ligand forms some sort of 'chiral pocket', like an enzyme active site, with the osmium sitting at the bottom of it. Alkenes can only approach the osmium if they are correctly aligned in the chiral pocket, and steric hindrance forces the alignment shown in the scheme above. The analogy with an enzyme active

site goes even further, since it appears that part of the pocket is 'attractive' to aromatic or strongly hydrophobic groups. This part appears to accommodate R_L , part of the reason why the selectivity in the dihydroxylation of *trans*-stilbene is so high.

The asymmetric dihydroxylation is much less fussy about the alkenes it will oxidize than Sharpless' asymmetric epoxidation. Osmium tetroxide itself is a remarkable reagent, since it oxidizes more or less any sort of alkene, electron-rich or electron-poor, and the same is true of the asymmetric dihydroxylation reagent. The following example illustrates both this and a synthetic use for the diol product.

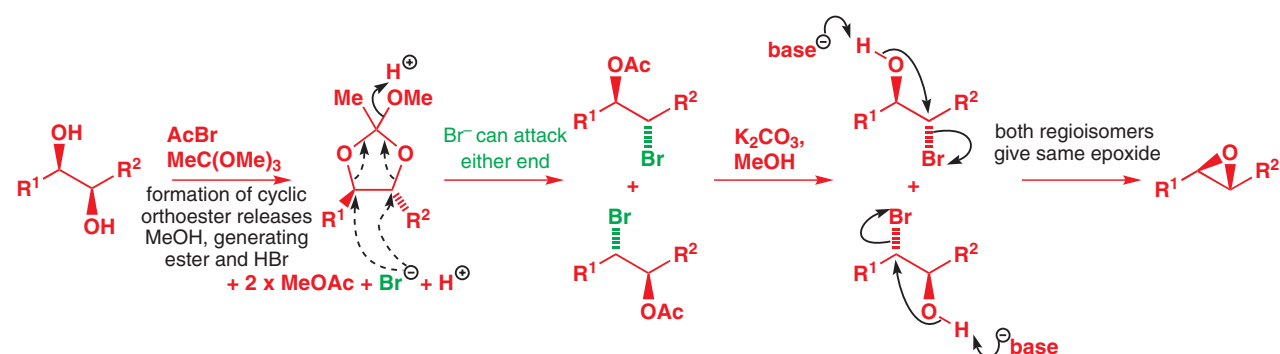


The chemists at Lilly in Spain who made this diol wanted to turn it into the protected amino acid shown after the dotted arrow as part of the synthesis of an anti-HIV compound. The ease with which diols can be made means that there are a number of reliable methods for transforming them into derivatives which undergo the sort of substitution needed. The one used here was to make the diol into a cyclic sulfate using sulfuryl chloride, SO_2Cl_2 . Cyclic sulfates behave like epoxides, and this one opens easily with azide at the more reactive position adjacent to the carbonyl group. Hydrolysis of the remaining sulfate ester, hydrogenation of the azide to the amine, and protection with Boc gave the target compound.



An alternative way of achieving the same transformation to the cyclic sulfate is to use thionyl chloride ($SOCl_2$) to give a sulfite, followed by ruthenium-catalysed oxidation to the sulfate.

Diols can even be converted with retention of stereochemistry directly to epoxides. Treatment of a diol with trimethyl orthoacetate and acetyl bromide gives firstly the cyclic orthoester, which opens with bromide to a regioisomeric mixture of the bromoacetates. The regiochemistry is irrelevant because treatment with base hydrolyses the ester and closes both of the resulting bromoalcohols to the same epoxide.

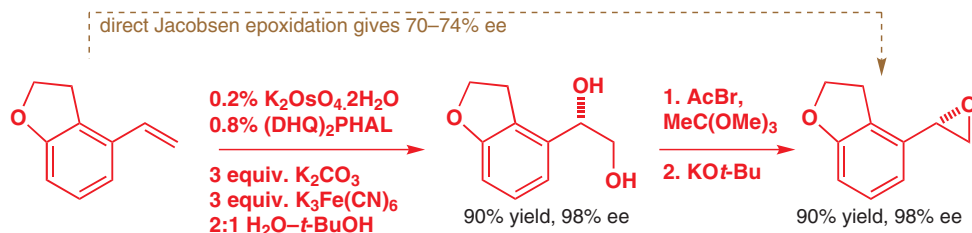


■ You can account for this by considering the mechanism of the dihydroxylation reaction (p. 905): it's a cycloaddition, so either the LUMO or the HOMO of the alkene can be involved.

■ We explained in Chapter 15 that S_N2 reactions adjacent to carbonyl groups are very fast. The regioselectivity of the ring opening of a cyclic sulfate, like that of an epoxide, is directed by the competition between relative rates of two nucleophilic substitution reactions. Benzylic and carbonyl-substituted positions usually open faster. There is more discussion of the regioselectivity of epoxide opening on p. 351.

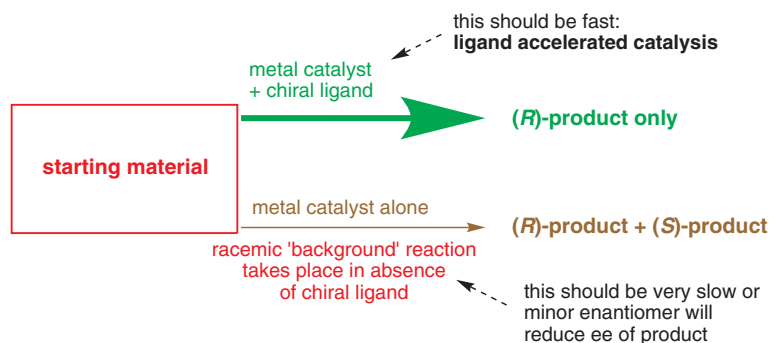
➡ See Chapter 39, p. 1059, for more on orthoesters.

It's no surprise that when chemists from Bristol Myers Squibb needed the epoxide below, they turned to asymmetric dihydroxylation rather than either of the epoxidation methods we have shown you. Sharpless epoxidation works only with allylic alcohols, and Jacobsen epoxidation performs poorly here, giving only 70–74% ee (mainly because the substrate is not a *cis* alkene). However, asymmetric dihydroxylation saves the day with 98% ee and around 90% yield, and a variant of the reaction we have just shown you gives the epoxide, also in 90% yield—well worth the extra step.



Ligand-accelerated catalysis

Asymmetric dihydroxylation is such a good reaction not just because of the careful way in which the ligands have been designed. It is a good reaction for a more fundamental reason: the reaction on which it is based (osmium-catalysed dihydroxylation) works only very poorly in the absence of the amine ligand. The chiral amine ligands don't just provide a chiral environment, they accelerate the reaction at the same time. This is what we mean by 'ligand accelerated catalysis'.



In any asymmetric reaction, we want the reagents to combine with one another only in the presence of the asymmetric influence provided by the chiral ligands. If the reaction works anyway, even without the chiral ligands, we have an uphill struggle because the reagents are quite capable of producing racemic product on their own. Racemic 'background' reactions are the reason why many of the reactions you are familiar with because they work so well racemically—addition of Grignard reagents to aldehydes, for example—don't really have good asymmetric versions. In the next section you will meet some more examples of reactions which are significantly accelerated by the presence of a chiral ligand.

■ In this section we will not consider the detailed mechanisms by which the stereochemistry of the ligand controls the stereochemistry of the product: in many cases this is not known anyway. We simply want to show you how the idea of ligand-accelerated catalysis has led to the discovery of new asymmetric reactions.

Asymmetric formation of carbon–carbon bonds

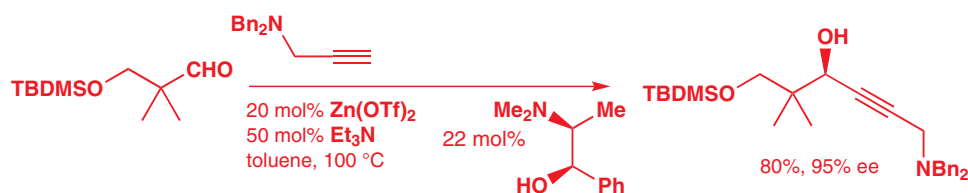
One of the first reactions you met in this book (Chapters 6 and 9) was the addition of an organometallic reagent to an aldehyde or a ketone. If the products of such an addition are chiral they are of course racemic. How might we make such a reaction enantioselective? One way would be to exploit the idea of ligand-accelerated catalysis and use a reaction which really doesn't work very well in the racemic series.

This is the case when the organometallic reagent is a dialkylzinc. Diethylzinc is commercially available as a solution in toluene or hexane, but it reacts only very slowly with an aldehyde

if the two are just left to stir together. However, if a chiral amino alcohol is added, the reaction becomes much faster. The amino alcohol forms a zinc alkoxide in the reaction mixture, and the coordination of this anionic chiral ligand to zinc both accelerates the transfer of zinc's alkyl groups to the aldehyde and makes that transfer enantioselective.



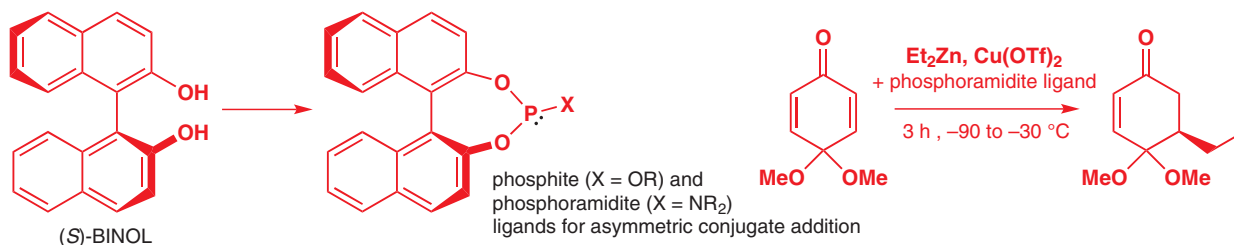
With alkyne nucleophiles, this reaction works with just catalytic amounts of zinc because the alkynylzinc forms in the reaction mixture when a weak base is added. It's a good way of making alkyne-containing alcohols.



Interactive mechanism for catalytic enantioselective organozinc addition to aldehydes

Asymmetric conjugate addition

In Chapter 22 we discussed the fact that copper promotes conjugate addition to electron-deficient double bonds. We can again exploit the low reactivity of organozinc compounds with carbonyl compounds, and with alkenes, if we add a catalytic amount of copper and a chiral phosphorus-containing ligand based on the atropisomeric binaphthyl structure you saw in BINAP. The organozinc has to transmetallate to the organocopper in order to react, and the copper always remains bound to the chiral ligand. Conjugate addition can only take place in a chiral environment, and good ees result.

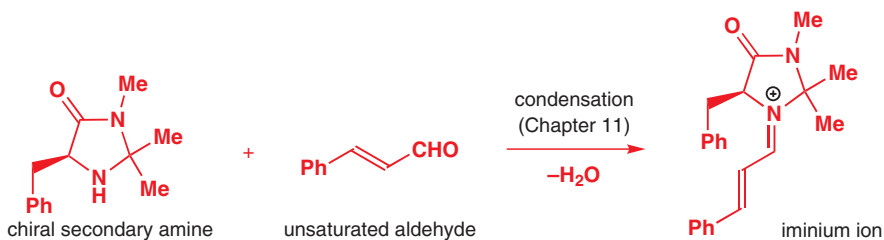


Organocatalysis

It will not have escaped your notice that most of the reactions we have presented in this chapter have made use of metals. Metals have labile coordination sites that can carry chiral ligands at the same time as they allow substrates and reagents to meet together in a chiral environment and then let the products dissociate so that the catalytic cycle can proceed. But in the early years of the 21st century, several chemists around the world realized that it is not always necessary to use a metal to initiate high levels of enantioselectivity in catalytic reactions. Simple chiral and enantiomerically pure organic molecules, many of them amines, can also react reversibly with substrates, providing a chiral environment and simultaneously activating them towards enantioselective attack.

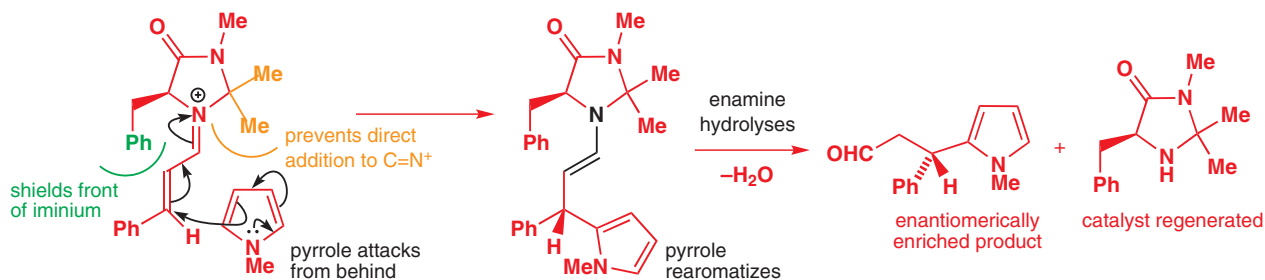
Here is an example, which picks up where we left off: a catalytic enantioselective conjugate addition. As you know from Chapter 11, aldehydes and ketones react with secondary amines to form enamines, via iminium ions. But this unsaturated aldehyde can't form an enamine because the iminium ion that is generated by condensation with the cyclic secondary amine

cannot lose a proton. The iminium ion is the end of the line for this condensation: it is very reactive towards attack by water (which would reversibly regenerate starting materials), but also towards attack by other nucleophiles. We have just what we want for good asymmetric catalysis—an intermediate species that is reactive, chiral, and enantiomerically pure.

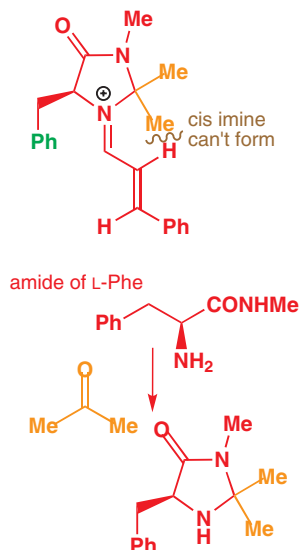


If this condensation is done in the presence of a weak nucleophile—strong enough to attack the positively charged iminium ion but not strong enough to attack the aldehyde itself—an addition reaction takes place. A pyrrole will do: pyrroles react well with cations. The phenyl ring highlighted in green hangs over the front of the molecule so the pyrrole has no choice but to attack diastereoselectively, from behind. The product is an enamine, which in the acid conditions of the reaction is hydrolysed by the water generated in the initial condensation, revealing the aldehyde in enantiomerically enriched form (93% ee) and regenerating the secondary amine catalyst.

➡ See p. 733 of Chapter 29 for some reactions of pyrrole. You should certainly look back to Chapter 29 if you need reminding why pyrrole reacts in its 2-position.



Interactive mechanism for enantioselective organocatalytic Friedel–Crafts alkylation



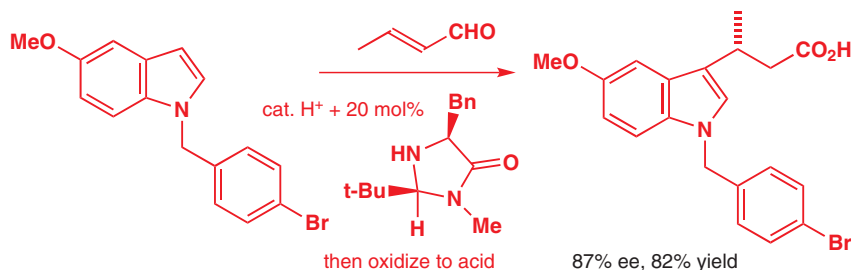
This catalyst and strategy were invented by the Glaswegian chemist David MacMillan at the California Institute of Technology (now at Princeton in New Jersey) and given the name 'organocatalysis'. Organocatalysis makes use of small organic molecules to achieve catalytic asymmetric transformations, and can be distinguished from the more widespread methods of catalysis which typically use metals. We'll introduce another type of organocatalysis towards the end of the chapter, but before we move on it's worth looking at this amine catalyst and the way it works in a little more detail.

The geminal dimethyl group highlighted in orange above is also important to the functioning of the catalyst. Without it, there is clearly a danger that the pyrrole will add directly to the C=N bond of the iminium ion, a reaction that would kill the catalyst because the product is an amine and not an enamine. The methyl groups on both faces of the iminium C=N bond stop this happening. The other thing it ensures is the geometry of the iminium C=N bond. This bond is *trans* so that the alkene can keep away from the quaternary carbon bearing the two orange groups; the benzyl group with the green phenyl may be bigger in terms of total number of atoms, but there is more space for the alkene on that side because the nearest carbon also carries just an H atom. Why is the geometry of the imine important? Well, if any of it were *cis*, it would present the other face to the pyrrole and would be likely to give the opposite enantiomer of product.

Catalysts aren't used in such great quantities as chiral auxiliaries, and so in general their synthesis does not need to be quite so direct. Nonetheless as you can see from the examples here, organocatalysts are still generally used in much greater quantities (10–20 mol%) than some of the best metal catalysts. In this case you should be able to spot that the left-hand

portion of the cyclic amine is a derivative of L-phenylalanine. Condensation of its *N*-methyl amide with an equivalent of acetone gives the catalyst itself.

Here's a related catalyst—as with the Rh- and Ru-catalysed reactions, fine tuning of the catalyst is important—being used in the synthesis of an important pharmaceutical compound, a COX-2 inhibitor. This time the nucleophile is an indole reacting characteristically at its 3-position.



Interactive mechanism for enantioselective organocatalytic indole alkylation

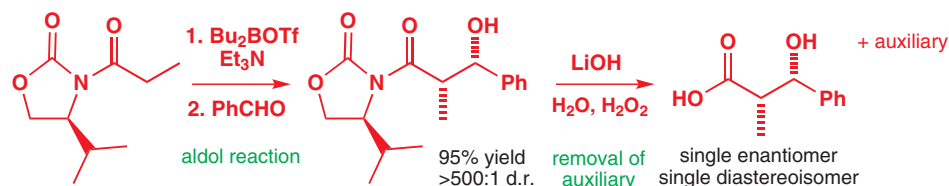
Asymmetric aldol reactions

You saw in Chapter 33 that it is possible to use aldol reactions to create two new chiral centres in a single step, and that the relative stereochemistry of the two chiral centres depends in many cases on the geometry of the enolate used to do the aldol reaction. The power of an *asymmetric* aldol reaction is easy to see: it creates two new chiral centres with control over their absolute stereochemistry, and also constructs a new C—C bond. What is more, the products of aldol reactions are very common features in a huge number of natural products known as polyketides—as you will see in the next chapter, polyketides are made by living things using a series of successive enzyme-controlled aldol reactions.

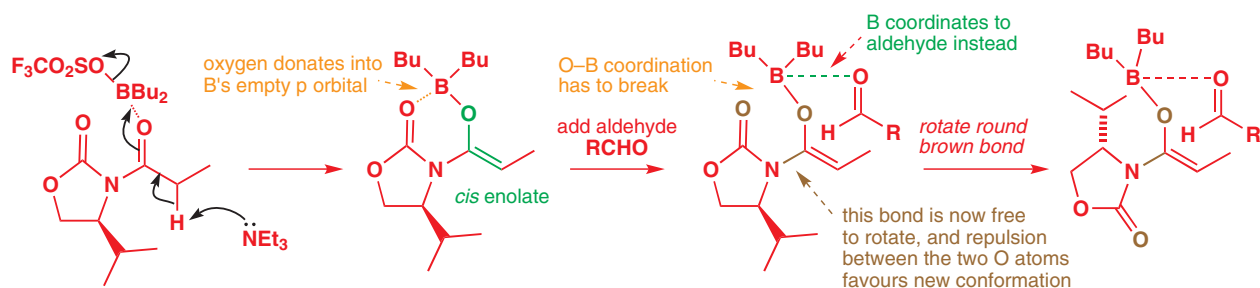
Chiral auxiliary-controlled aldol reactions: the Evans aldol

An aldol reaction is the addition of an enolate to an electrophile, where the electrophile is an aldehyde or a ketone. You have already seen earlier in this chapter how enolates can be used to make new C—C bonds enantioselectively when we explained how to control enolate alkylation with Evans' chiral auxiliaries. Evans' auxiliaries also provide one of the most straightforward ways of carrying out asymmetric aldol reactions, and we will start with an example before explaining how asymmetric aldol reactions can be done using catalytic methods.

This aldol reaction is carried out using a base (triethylamine) and dibutylboron triflate plus benzaldehyde. The aldol product is formed with outstandingly good selectivity and in high yield, and all that remains is to remove the auxiliary with base and isolate the hydroxy-acid as a single diastereoisomer and a single enantiomer.



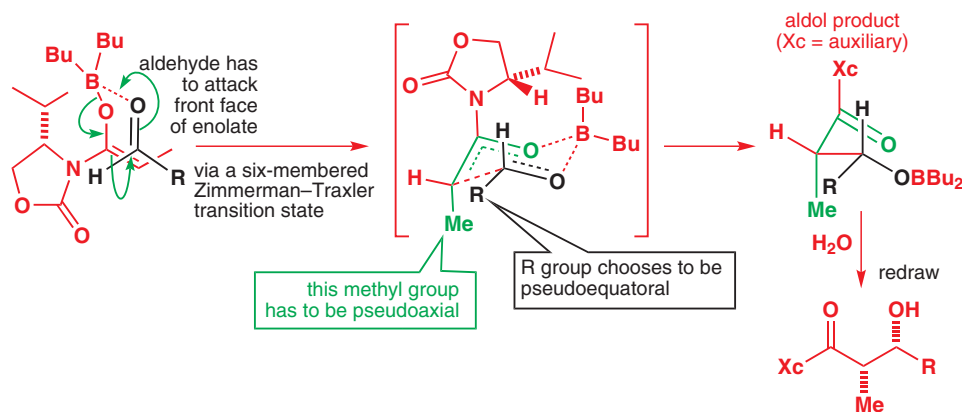
Aldol reactions using the lithium enolate of the acylated auxiliary shown here fail to give good selectivities, so instead we use the boron enolate. The combination of triethylamine and the boron triflate form the stable boron enolate, which has to be *cis* because the size of the auxiliary prevents the *trans* enolate forming. Boron has an empty p orbital, and donation into this orbital from the oxygen of the carbonyl group stabilizes the enolate.



Now the aldehyde is added. If the reaction is to take place, the aldehyde must coordinate to the boron because boron enolates aren't reactive enough to attack aldehydes unless they are activated by coordination to a Lewis acid. However, the aldehyde can't simply coordinate to the boron atom of the enolate because then the boron will end up with five bonds, which is impossible for a first-row element. So, if the reaction is to continue, the boron has to let go of the auxiliary's carbonyl group and coordinate to the aldehyde instead.

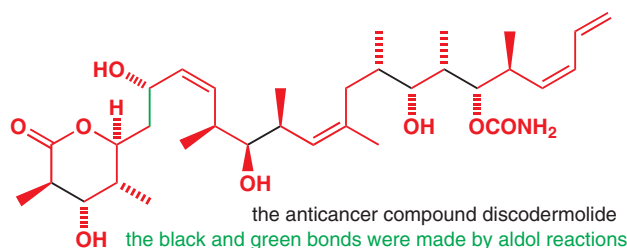
At this stage something rather remarkable happens: now that the boron is no longer holding the two oxygen atoms of the enolate close together, repulsion between them (they are both electron-rich atoms) forces the auxiliary part of the enolate to swing round through 180° and end up pointing in the opposite direction. This is highly significant for what happens next because you can see from the last structure in the scheme above that this rotation ends up swinging the isopropyl group of the auxiliary round to the underside of the enolate and therefore forcing the auxiliary to react from the front instead of the back.

The diagrams below continue the story. The aldehyde has to attack the front face of the auxiliary, but it also has to do so through what we termed in Chapter 33 a 'Zimmerman-Traxler transition state'—a six-membered, chair-like cyclic structure which allows the enolate to attack the aldehyde while simultaneously transferring the metal (here the boron) from the enolate oxygen to the new hydroxyl group.



Interactive mechanism for the chiral auxiliary-controlled aldol reaction

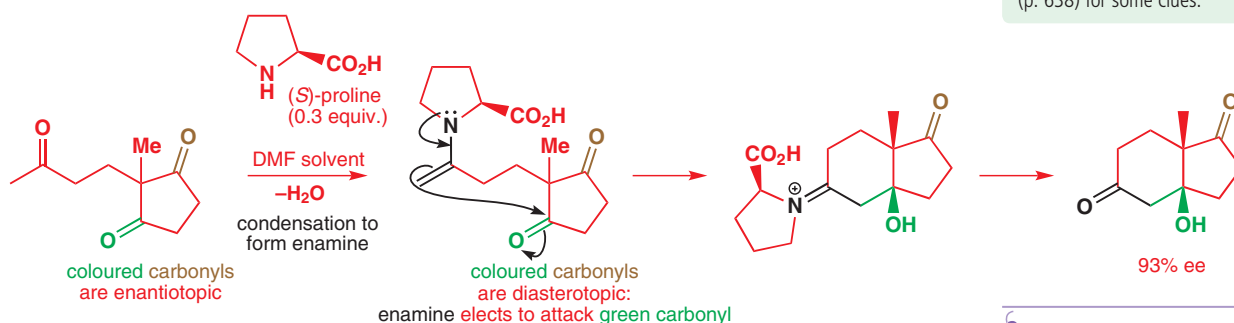
All the usual advantages and disadvantages of chiral auxiliaries apply here: the products are formed in very high selectivity and can be purified to high ee, but the extra steps required to introduce and remove the auxiliary may compromise the overall yield and efficiency of the reaction. Nonetheless, using this method and many others like it, it has now become possible routinely to make polyketide natural products by successive aldol reactions, mimicking nature's approach to these compounds. In a spectacular demonstration of the power of synthetic chemistry to outperform natural sources, chemists at the Swiss pharmaceutical company Novartis made 60 g of the anticancer compound discodermolide by a synthetic route, including five aldol reactions. Four of these, shown by black C-C bonds in the structure below, used the aldol or alkylation reactions controlled by Evans' oxazolidinone chiral auxiliaries. To obtain the same quantity from the natural source, the sponge *Discodermia*, is impossible: the sponge produces minuscule amounts of discodermolide and can be harvested only by using manned submersible vehicles, at some cost to the marine environment.



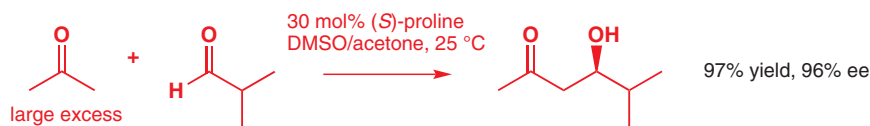
Aldol reactions catalysed by proline

A variety of catalytic ways of doing asymmetric aldol reactions have also been invented, but space prevents us discussing all but one. This one we highlight firstly because it illustrates the use of a supremely simple biologically derived compound to catalyse a complex reaction, and secondly because this discovery was part of the revolution in catalytic thinking which launched the field of organocatalysis in the early years of the 21st century.

The catalyst we will use is the amino acid L-proline—no derivatization or protection required. It was actually back in 1971 that it was first noted that L-proline will catalyse asymmetric aldols, but until the year 2000 examples were limited to this one cyclization. Treatment of a triketone with proline leads to selective cyclization onto one of the two enantiotopic carbonyl groups. A molecule of proline must condense with the least hindered ketone, and in this case an enamine (rather than an iminium ion) can form. The chiral enamine can select to react with only one of the two other carbonyl groups, and it turns out that it chooses with rather high selectivity the one coloured green in the scheme below. Cyclization, in the manner of a Robinson annelation, and hydrolysis of the resulting iminium ion follow on, releasing the molecule of L-proline to start another catalytic cycle. The isolated product is the bicyclic ketone, in 93% ee.



This remained an oddity of a reaction until 2000, when chemists at the Scripps Institute in California, and then others around the world, took the simple expedient of adding L-proline to many other aldol reactions, with considerable success. With care, excellent results can be obtained. Here is an example.



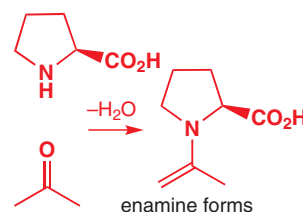
You will remember from Chapter 26 that crossed aldol reactions between enolizable partners, like these, usually need one of the reagents to be converted to an enolate equivalent to ensure selective reaction. Here, the acetone is in excess, but the components are just stirred together at room temperature in DMSO! The key to success is that one of the two components must be more able to form a reactive enamine with proline than the other. In the case above, the acetone-derived enamine is favoured because (1) enamine formation is reversible, (2) the acetone is in excess, and (3) the enamine from acetone is less hindered and more reactive than the enamine that would arise from the aldehyde.

➡ You saw the racemic example of this Robinson annelation on p. 638.

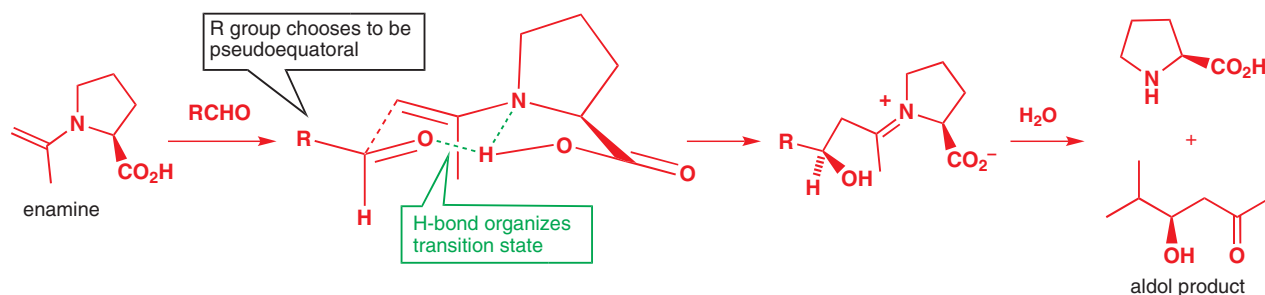
■ You might like to think about how the starting material for this reaction can be made. Look back at the section on the Robinson annelation in Chapter 26 (p. 638) for some clues.

🔗 Interactive mechanism for the proline-catalysed Robinson annelation

■ It has an odd name too: the cyclization is sometimes called the Hajos–Parrish–Eder–Sauer–Wiechert reaction, after its discoverers, but only by those who want to impress their friends.



In the aldol reaction itself, proline's carboxyl group has a key role to play because it can participate in a hydrogen bond that organizes the six-membered transition state in such a way that only one of the possible enantiomeric products can form. The diagram below shows how. Water generated in the initial condensation hydrolyses the iminium product of the aldol and regenerates the proline catalyst.

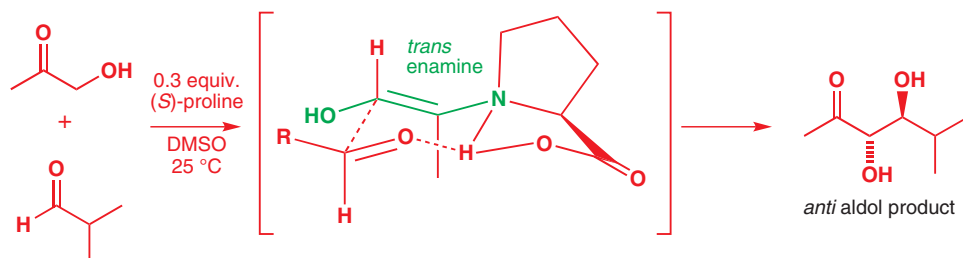


Interactive mechanism for the proline-catalysed enantioselective aldol reaction

➡ See p. 868 for a discussion of enolate geometry and *syn* and *anti* aldols.

Interactive mechanism for the proline-catalysed *anti* selective aldol reaction

Organocatalytic aldol reactions also work well with hydroxylated ketones—the reaction below, for example. In this case, the enamine forms with an *E* double bond, which means that the hydroxyl group has to be equatorial on the six-membered transition state. You should be able to work out from this that the *anti* aldol has to form.



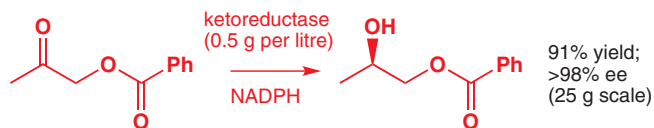
Enzymes as catalysts

We pointed out at the beginning of the chapter that all enantiomeric purity must ultimately derive from nature. We have almost come full circle: the reactions we have just been looking at use one of nature's protein building blocks, L-proline, directly as a catalyst. Even more intriguingly, the reaction just above, which forms a ketodiol, is extremely reminiscent of the aldol reactions which nature uses to build carbohydrates, as you will see in the next chapter.

Yet nature does not use single amino acids to catalyse asymmetric reactions, it uses enzymes. Enzymes are vastly more efficient than L-proline and catalyse a much wider range of reactions, but while they are also much more complicated, their reactivity derives ultimately from the amino acids they are made up of.

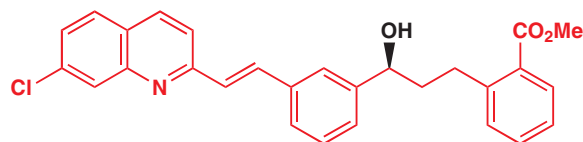
Life uses enzymes to catalyse asymmetric reactions, so the question is—can chemists? The answer is yes, and there are many enzymes that can be produced in quantities large enough to be used in the catalytic synthesis of enantiomerically pure molecules. This field—known as biocatalysis—melds ideas in chemistry and biology, and we do not have the space here to discuss it in detail. We leave you with just one example: the reduction of a ketone to an alcohol with an enzyme known as a ketoreductase.

➡ The structure of NADPH is on p. 1150.



The ketoreductase takes hydride from the reducing agent NADPH (which you will meet in the next chapter) and transfers it enantioselectively to the carbonyl group in the active site of

the enzyme. This ketoreductase, isolated from yeast, may never have met this non-biological substrate—benzoyloxyacetone—before, but the reaction works. In fact this sort of reaction works so well that ketoreductases are used to carry out the reduction needed to produce the pharmaceutical intermediate discussed on p. 1116 on a 230 kg scale.



made by enzymatic reduction on a 1/4-tonne scale

Many other groups of enzymes behave similarly: they have evolved to take part in particular biochemical pathways, but they are sufficiently promiscuous that they will happily accept alternative substrates and provide chemically useful products from them. Enzymes are catalysts, like any other. In the next chapter, we take a more detailed look at those biochemical pathways and discuss the organic chemistry of life.

● Summary of the main methods for asymmetric synthesis

Method	Advantages	Disadvantages	Examples
resolution	both enantiomers available	maximum 50% yield	synthesis of BINAP
chiral pool	100% ee usually guaranteed	often only one enantiomer available	amino acid and sugar derived syntheses
chiral auxiliary	often excellent ees; can recrystallize to purify to high ee	extra steps to introduce and remove auxiliary	oxazolidinones
chiral reagent	achieve some otherwise difficult transformations	only a few reagents are successful and often for few substrates	alkyllithium-(–)-sparteine complex
chiral catalyst	economical: only small amounts of recyclable material used	only a few reactions are really successful; recrystallization can improve only already high ees	asymmetric hydrogenation, epoxidation, dihydroxylation

Further reading

For an overview of the relationship between smell and stereochemistry, see R. Bentley, *The Nose as a Stereochemist: Enantiomers and Odour*, *Chem. Rev.*, 2006, **106**, 4099. Interesting examples of the use of asymmetric methods in the large scale synthesis of drug molecules are given in M. Ikunaka, *Chem. Eur. J.* 2003, **9**, 379. The prevalence of chiral drugs and the relative importance of asymmetric synthesis and resolution are discussed in B. Kasprzyk-Hordern, *Chem. Soc. Rev.*, 2010, **39**, 4466 and in J. S. Carey, D. Laffan, C. Thomson and M. T. Williams *Org. Biomol. Chem.* 2006, 2337.

For a more advanced treatment of asymmetric synthesis see P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapters 22–31 and the accompanying *Workbook*, also Wiley, 2008. Very detailed advanced mechanistic discussion of asymmetric hydrogenation and asymmetric epoxidation methods may be found in *Asymmetric Synthesis*, vol. 5, ed. J. D. Morrison, Academic Press, New York (1985). This five-volume set is now rather dated but provides some very valuable discussion of the

classic methods of asymmetric synthesis. There are reviews of asymmetric dihydroxylation in *Chem. Rev.* 1994, **94**, 2483 and *Org. Synth.* 1996, **73**, 1 and of asymmetric hydrogenation in *Acc. Chem. Res.* 2007, pp. 1237–1419 and the *Handbook of homogeneous hydrogenation* published by Wiley, 2007. A very recent comprehensive overview of asymmetric methods can be found in the multi-volume set *Comprehensive Chirality*, pub. Elsevier, 2011.

The Cilag resolution of the pyridyl amino acid is described in *Org. Process Res. Dev.* 2001, **5**, 23. For an informative comparison of different auxiliary and catalytic methods for the synthesis of a simple chiral carboxylic acid, see *Org. Process Res. Dev.* 2003, **7**, 370. For a leading reference to the use of enzymes to reduce ketones, see the account of the Codexis work on montelukast in *Org. Process Res. Dev.* 2010, **14**, 193. The spectacular synthesis of discodermolide by Novartis using a series of aldol reactions is described in *Org. Process Res. Dev.* 2004, **8**, 92, 101 and 107.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>